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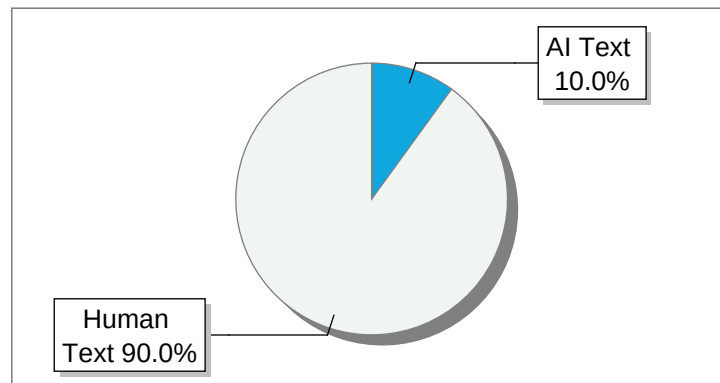
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VISVESVARAYA TECHNOLOGICAL UNIVERSITY “Jnana Sangama”, Belagavi-590018, Karnataka Report on Neuro Adaptive AI Agents A Neuroplasticity Framework for Adaptive Multimodal Reasoning Submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science & Engineering Submitted by USN Name 1BI22CS152 Shashi Chidanand Madari 1BI22CS133 Rohit Kishan Honakhande 1BI22CS164 Srivallabha K M 1BI23CS400 Abhishek Srivatsav Under the Guidance of Prof.

KN Prashanth Kumar Assistant Professor, Department of CS&E, BIT Bengaluru-560004 DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING BANGALORE INSTITUTE OF TECHNOLOGY KR. Road, VV.

Pura, Bengaluru-560 004 2025-26 VISVESVARAYA TECHNOLOGICAL UNIVERSITY “Jnana Sangama”, Belagavi-590018, Karnataka BANGALORE INSTITUTE OF TECHNOLOGY Bengaluru-560 004 DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING Certificate This is to certify that the project work entitled Neuro Adaptive AI Agents A Neuroplasticity Framework for Adaptive Multimodal Reasoning carried out by USN Name 1BI22CS152 Shashi Chidanand Madari 1BI22CS133 Rohit Kishan Honakhande 1BI22CS164 Srivallabha KM 1BI23CS400 Abhishek Srivatsav Bonafide students of VII semester BE. for the partial fulfillment of the requirements for the Bachelors Degree in Computer Science & Engineering of the VISVESVARAYA TECHNOLOGICAL UNIVERSITY during the academic year 2025-26.

It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library.

The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said degree.

Prof KN Prashanth Kumar Assistant Professor Dr.

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Shanthala S Principal, BIT External Viva Name of the Examiners Signature with date i VISVESVARAYA TECHNOLOGICAL UNIVERSITY “Jnana Sangama”, Belagavi-590018, Karnataka BANGALORE INSTITUTE OF TECHNOLOGY Bengaluru-560 004 CERTIFICATE Certified that the project work entitled Neuro Adaptive AI Agents A Neuroplasticity Framework for Adaptive Multimodal Reasoning carried out by Mr/Ms.

Shashi Chidanand Madari, Rohit Kishan Honakhande, Srivallabha K M, Abhishek Srivatsav, Bearing the USN 1BI22CS152, 1BI22CS133, 1BI22CS164, 1BI23CS400 respectively a bonafide student of COMPUTER SCIENCE AND ENGINEERING in partial fulfillment for the award of Bachelor of Engineering / Bachelor of Technology in BANGALORE INSTITUTE OF TECHNOLOGY of the Visvesvaraya Technological University, Belgaum during the year 2025-2026 It is certified that, he/she has completed the project satisfactorily.

Prof KN Prashanth Kumar Dr.

Suneetha K R Assistant Professor, CSE Prof.

& Head Dept.

of CSE ii ACKNOWLEDGEMENT The knowledge & satisfaction that accompany the successful completion of any task would be incomplete without mention of people who made it possible, whose guidance and encouragement crowned our effort with success.

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Shashi Chidanand Madari Rohit Kishan Honakhande Srivallabha KM Abhishek Srivatsav iii ABSTRACT This project presents a Neuro-Adaptive AI Agent Framework inspired by principles of neuroplasticity to achieve robust and context-aware multimodal reasoning across image, audio, and text inputs.

The system employs three specialized agents Normal, Blind, and Mute each designed to compensate dynamically for missing or degraded sensory modalities.

A custom 15- dimensional latent embedding space enables alignment of heterogeneous inputs through modality-specific encoders and cross-modal fusion.

The architecture integrates vision captioning, speech and text understanding, affect- aware features, and causal reasoning, supported by a cognitive memory retrieval mechanism.

Interactive 3D and 2D semantic visualizations demonstrate adaptive reasoning behavior, semantic alignment, and controlled drift across agents.

Experimental results show that the system reorganizes reasoning pathways dynamically without retraining, highlighting its potential as a resilient, interpretable, and neuro-inspired approach toward generalized artificial intelligence.

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31 Software tools 22-23 32 Hardware tools 23-24 41 Project Development Phases 33-35 42 Team Role Allocation 35-36 71 Modality Input Testing 74 72 Encoder Projection Testing 75 73 Query Builder Testing 75 74 Memory Storage Testing 75 75 Memory Retrieval Testing 76 76 Agent Reasoning Testing 76 77 PCA Visualization Testing 76 78 Accuracy Assessment of PCA-Based Modality Visualization 77 79 Multimodal Fusion Pipeline Accuracy 78 710 Memory–Reasoning Integration Accuracy 79 711 Cognitive Compression Accuracy 79 712 Cross-Agent Interaction Accuracy 80 713 PCA Visualization Accuracy 80 714 End-to-End System Accuracy 81 81 Observed Centroid Coordinates of Input Embeddings 83 82 Accuracy Comparison Summary 88 83 Reasoning Proximity Analysis Across All Inputs 93 CHAPTER 1 INTRODUCTION Dept. of CS&E, BIT 2025-26 1 CHAPTER 1 INTRODUCTION Artificial Intelligence systems were initially developed to process information from a single type of input such as text, image, or audio.

However, real-world problems often involve multiple forms of data, and human intelligence naturally combines information from different senses.

Humans also adapt their reasoning when one or more sensory inputs are unavailable.

For example, visually impaired individuals rely more on hearing, while people with speech impairments depend on visual and textual information.

This adaptive behavior motivates the development of AI systems that can adjust their reasoning process based on available inputs.

This project presents a Neuro-Adaptive Multi-Agent Intelligence System inspired by the concept of neuroplasticity, where adaptability is achieved through alternative reasoning strategies rather than biological modeling.

In this system, neuroplasticity is represented as the ability of artificial agents to modify their reasoning behavior depending on the presence or absence of specific input modalities.

The system follows a multi-agent architecture consisting of three agents □ Normal Agent, which performs multimodal reasoning using image, audio, and text inputs □ Blind Agent, which operates without visual input and relies on audio and text □ Mute Agent, which functions without audio input and uses image and text The proposed system integrates deep learning–based feature extraction, multimodal representation, and memory-supported reasoning.

Inputs from image, audio, and text modalities are encoded into a shared 15-dimensional latent embedding space, allowing comparison and retrieval across modalities.

Image information is processed using a ResNet-based encoder and captioning mechanism, audio information is processed through speech detection, speech-to-text conversion, and a CNN-based encoder, while text information is represented using SBERT embeddings.

A memory module stores previously processed multimodal information along with contextual importance and emotion-related metadata.

During reasoning, relevant past entries are retrieved to support the current decision process, improving contextual consistency and explanation quality.

This mechanism provides an artificial analogy to Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 2 how experience supports human decision-making, without attempting to replicate biological processes.

The system includes a reasoning and explanation module that generates structured outputs based on agent type. Explanations are provided in text or audio form depending on the sensory constraints of the agent.

In addition, the system offers 3D visualization of multimodal input embeddings and reasoning outputs using Principal Component Analysis (PCA), enabling clear observation of how different agents reason under varying input conditions.

Overall, this project demonstrates how neuro-adaptive behavior in AI systems can be practically implemented using a multi-agent framework, multimodal embeddings, memory-based reasoning, and adaptive explanation

strategies suitable for real-world applications.

11 Overview The rapid evolution of Artificial Intelligence has led to systems that can perceive, analyze, and reason over complex data.

However, most intelligent systems are still designed around fixed input assumptions, where the availability of complete and reliable data is taken for granted.

In practical environments, this assumption rarely holds true.

Sensory inputs such as images, speech, or text may be incomplete, noisy, or entirely absent due to environmental, technical, or human-related constraints.

Human cognition demonstrates a fundamentally different approach.

Instead of failing when sensory information is missing, the human brain adapts by reallocating cognitive focus to the available senses.

This adaptability allows humans to continue reasoning, learning, and decision-making under constrained conditions.

Translating this adaptive behavior into artificial systems remains a significant research challenge in multimodal intelligence.

This project explores an engineering-oriented solution to this challenge by introducing a neuro-adaptive reasoning framework implemented through a coordinated multi-agent architecture.

Rather than building a single monolithic model, the system distributes reasoning responsibilities across specialized agents, each designed to operate under specific sensory conditions.

This design choice emphasizes robustness, flexibility, and interpretability over raw predictive performance.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 3 The system combines multimodal feature extraction, unified representation learning, and memory-supported reasoning to create an adaptive inference pipeline.

Instead of treating missing modalities as failure cases, the system treats them as contextual constraints that guide agent selection and reasoning strategy.

This shift enables the system to maintain functional reasoning even when inputs are partially unavailable.

By integrating memory-driven context retrieval, the system further enhances reasoning continuity across interactions.

Past experiences are reused to improve interpretation quality, enabling consistent explanations rather than isolated responses.

Visualization components are incorporated to make internal representations observable, supporting transparency and academic analysis of system behavior.

In summary, this project provides a structured investigation into adaptive multimodal reasoning using artificial agents.

It focuses on practical system design, explainability, and robustness, making it suitable as a foundational study for neuro-inspired AI systems in academic and applied domains.

12 Objectives The main objective of this project is to design and implement a multimodal intelligent system that can adapt its reasoning process based on the availability of sensory inputs.

The system aims to simulate adaptive human reasoning using multiple AI agents and memory-supported decision making.

121 Core System Objectives □ To design a multimodal AI system capable of processing image, audio, and text inputs in a unified manner.

□ To develop specialized AI agents that can operate effectively even when one sensory modality is unavailable.

□ To enable adaptive reasoning by allowing agents to compensate for missing inputs using the available modalities.

□ To implement a shared low-dimensional embedding space to represent multimodal data in a consistent and comparable format.

□ To design a memory module that stores previous experiences and retrieves relevant information to support current reasoning tasks.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 4 122 Technical Model Objectives □ To implement deep learning-based encoders for extracting meaningful features from different input modalities.

- To represent image, audio, and text data using a common latent feature space to support similarity comparison and retrieval.
- To integrate emotion-aware processing so that emotionally significant inputs are given higher importance during memory storage and retrieval.
- To generate structured and context-aware reasoning outputs based on retrieved memory and current input information.
- To adapt explanation formats according to the agent type, ensuring accessibility under different sensory constraints.

123 Deployment Objectives □ To develop a web-based application using the Flask framework for uploading multimodal inputs and executing different agents.

□ To provide visual analysis of multimodal and reasoning embeddings using dimensionality reduction techniques such as PCA.

□ To ensure that the system is modular and scalable, allowing future extensions and improvements.

□ To demonstrate the practical applicability of adaptive multimodal AI systems through an interactive and user-friendly interface.

13 Purpose, Scope, and Applicability 131 Purpose The primary purpose of this project is to design and implement a multimodal intelligent system that can adapt its reasoning process based on the availability of sensory inputs such as image, audio, and text.

The system aims to demonstrate how artificial agents can adjust their behavior when one or more input modalities are missing, similar to how humans adapt their thinking using available senses.

The project focuses on □ Adaptive reasoning using multiple input modalities □ Use of memory to support decision-making □ Generating suitable explanations based on agent type □ Demonstrating practical multimodal AI concepts through an interactive system Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 5 This work is intended as an academic implementation to understand multimodal learning, memory-based reasoning, and adaptive agent behavior.

132 Scope The scope of this project includes the design, development, and demonstration of a multimodal AI system that integrates image, audio, and text processing with memory- supported reasoning.

The system is implemented using a modular architecture so that each component can be understood, tested, and extended independently.

1321 Multimodal Signal Processing □ Image Processing o Image caption generation using a pretrained vision-language model o Facial emotion detection to identify emotional context o Feature extraction using a ResNet-based image encoder o Projection of image features into a shared low-dimensional embedding space □ Audio Processing o Speech detection and speech-to-text conversion o Audio preprocessing including noise handling and normalization o Feature extraction from audio signals using a CNN-based encoder o Emotion detection from transcribed speech □ Text Processing o Semantic embedding of text using SBERT o Emotion detection from text input o Keyword and semantic similarity analysis o Generation of structured summaries for reasoning 1322 Cognitive Memory Framework □ Storage of multimodal experiences including image, audio, and text information □ Each memory entry contains o Encoded feature vectors o Semantic embeddings o Emotion and importance information o Contextual summaries □ Retrieval of relevant memories based on similarity, importance, and contextual Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 6 relevance □ Use of retrieved memories to enhance current reasoning and explanation generation 1323 Multi-Agent Adaptive Reasoning □ Normal Agent o Uses image, audio, and text inputs together o Provides complete multimodal reasoning output □ Blind Agent o Operates without visual input o Uses audio and text for reasoning o Produces audio-based explanations □ Mute Agent o Operates without audio input o Uses image and text for reasoning o Produces clear and concise text-based explanations □ Agent Interaction o Supports conversion of explanations between agents o Enables sharing of reasoning results in formats suitable for different agents 1324 Reasoning and Explanation Module □ Generation of structured reasoning outputs based on current inputs and retrieved memory □ Adaptation of explanation format depending on agent type □ Support for both text and audio explanations □ Use of contextual and emotional information to improve explanation relevance 1325 Visualization and Deployment □ Visualization o 3D visualization of multimodal input embeddings o Comparison of reasoning outputs across different agents o Helps in understanding modality contribution and adaptive behavior □ Deployment o Web-based interface developed using Flask o Supports

multimodal input uploads Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 7 o Allows selection and execution of different agents o Displays explanations, logs, and visualization outputs 133 Applicability This system has broad applicability across multiple AI, cognitive computing, and engineering domains.

Its neuro-adaptive architecture allows it to function effectively under sensory limitations, making it valuable in real-world multimodal environments.

□ Assistive Artificial Intelligence The system is particularly useful in assistive applications where users may have sensory impairments.

o Support for visually impaired users The Blind Agent can provide audio- based explanations by processing speech and text inputs, allowing visually impaired users to understand visual content through auditory descriptions.

o Support for speech-impaired users The Mute Agent produces structured text-based explanations using image and text inputs, enabling communication and learning without relying on speech.

These features improve accessibility and ensure inclusive human–computer interaction.

□ Cognitive Robotics Robotic systems require dynamic sensor fusion and compensation mechanisms when one sensor fails or produces incomplete data.

o Multisensory robot perception Robots can use adaptive multimodal embeddings to understand surroundings despite missing visual or audio input.

o Adaptive reasoning in dynamic environments The neuroplastic reasoning model allows robots to reformulate decisions when sensor channels degrade due to noise, lighting, or interference.

o Human–robot collaborative interaction Agents can interpret human communication through alternative modalities, improving collaborative safety and responsiveness.

□ Healthcare and Accessibility The emotion-aware and memory-supported reasoning framework makes the system suitable for healthcare-related applications.

o The system can adjust its explanation style according to user needs.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 8 o Emotion-aware processing can assist in monitoring user responses during interactions.

o It can be used as a supportive tool in therapy, patient assistance, and accessibility-focused applications.

□ Intelligent Surveillance and Monitoring The multimodal processing capability allows the system to be applied in monitoring and analysis tasks.

o Integration of image, audio, and text improves understanding of events.

o The system can generate summarized explanations from multimodal inputs.

o Useful in applications such as smart monitoring systems and event interpretation platforms.

□ Educational and Learning Applications The system can be applied in educational environments as an intelligent tutoring or support system.

o Provides explanations in text or audio based on user requirements.

o Supports adaptive explanation formats for better understanding.

o Useful for students with different learning preferences or limitations.

□ AI Education and Research The platform provides a controlled environment to study computational cognitive principles.

o Platform for studying cross-modal embedding learning Researchers can analyze how unified representation spaces align semantically different modalities.

o Simulation of neuroplasticity in engineered AI agents The Blind and Mute Agents demonstrate computational analogs of sensory compensation and adaptive reasoning.

o Experimental setup for multi-agent communication and reasoning The system supports inter-agent feedback loops, enabling study of cooperative and collaborative intelligence.

□ General Multimodal AI Applications Beyond specialized domains, the system generalizes across multiple multimodal Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 9 AI tasks.

o Multimedia analysis Capable of extracting, aligning, and interpreting information from videos, images, audio clips, and text o Context-aware virtual agent systems Personalized assistants can dynamically switch reasoning

strategies based on available modalities.

o Intelligent tutoring and reasoning platforms Neuro-adaptive reasoning allows automated tutors to generate explanations suited to user preferences and cognitive context.

14 Organization of the Report This report is organized into nine chapters, each describing a specific stage of the project development process.

The structure of the report is designed to present the project in a clear and systematic manner, starting from basic concepts and progressing toward implementation, results, and conclusions.

Chapter 1 Introduction This chapter introduces the project and provides an overview of the problem domain. It includes the background of the study, objectives of the project, purpose, scope, applicability, and the overall organization of the report.

Chapter 2 Literature Survey This chapter presents a review of existing work related to multimodal artificial intelligence systems, adaptive reasoning models, memory-based AI, and human– computer interaction. It highlights the limitations of current systems and motivates the need for the proposed solution.

Chapter 3 Requirement Engineering This chapter defines the requirements necessary for developing the proposed system.

It includes □ Hardware and software requirements □ User requirements □ System requirements □ Use case diagram □ Sequence diagram Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 10 □ Activity diagram Chapter 4 Project Planning This chapter explains the planning and scheduling of the project.

It includes project milestones, task distribution, timeline planning, and risk assessment to ensure smooth execution of the project.

Chapter 5 System Design This chapter describes the overall architecture and internal design of the system.

It includes □ System architecture diagram □ Module-wise design explanation □ Data flow between different components □ Design of agents, memory module, and reasoning module Chapter 6 Implementation This chapter discusses the implementation details of the project.

It explains how different modules are implemented, integrated, and executed.

The chapter also highlights the tools, libraries, and programming techniques used.

Chapter 7 Testing This chapter describes the testing strategies used to validate the system.

It includes □ Unit testing of individual modules □ Integration testing of the complete system □ Verification of agent behavior and system outputs Chapter 8 Results and Discussion This chapter presents the results obtained from the system.

It includes □ Outputs generated by different agents □ Visual analysis of multimodal and reasoning embeddings □ Discussion on system behavior under different input conditions Chapter 9 Conclusion and Future Work This chapter summarizes the work carried out in the project and highlights the keyNeuro Adaptive AI Agents Dept. of CS&E, BIT 2025-26 11 contributions.

It also discusses possible future enhancements such as improving reasoning accuracy, extending memory handling, and supporting additional modalities.

References This section lists the books, research papers, online resources, and tools referred to during the development of the project.

Annexures Includes supporting materials □ Annexure A Glossary of technical terms □ Annexure B Published research paper or acceptance letter □ Annexure C Project exhibition certificate or funding approval CHAPTER 2 LITERATURE SURVEY Dept.

of CS&E, BIT 2025-26 12 CHAPTER 2 LITERATURE SURVEY 21 Introduction Multimodal Artificial Intelligence aims to develop systems that can process and combine information from multiple input sources such as images, audio, and text.

While such systems improve contextual understanding, many existing approaches assume the availability of all input modalities at all times.

In real-world scenarios, this assumption is often unrealistic.

Inspired by the concept of neuroplasticity, where humans adapt their cognitive processes when a sensory modality is unavailable, this project adopts an adaptive reasoning approach.

The proposed system simulates neuroplastic behavior by dynamically selecting reasoning strategies based on the available modalities, without modifying the underlying model structure.

The system employs a neuro-adaptive multi-agent framework consisting of normal, blind, and mute agents, each designed to operate under specific sensory constraints.

The literature reviewed in this chapter discusses existing work related to multimodal learning, adaptive AI systems, memory-based reasoning, and agent architectures that collectively motivate the proposed approach.

22 Summary of Papers Paper 1 Multimodal Machine Learning Fusion, Alignment, and Co-Learning Authors Tadas Baltrušaitis, Chaitanya Ahuja, Louis-Philippe Morency Reference A.

Baltrušaitis et al, “Multimodal Machine Learning A Survey,” IEEE TPAMI, 2019.

Summary This work defines a structured taxonomy for integrating multiple data modalities, addressing five core challenges □ Representation – Designing joint feature spaces combining heterogeneous data (eg, image+text).

□ Translation – Generating one modality from another (eg, text-to-image).

□ Alignment – Establishing cross-modal relationships (eg, phoneme-to-image alignment) Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 13 □ Fusion – Merging modalities at different network stages.

□ Co-learning – Transferring knowledge across modalities (eg, learning from missing data).

The paper highlights a major limitation most systems assume complete modality availability, limiting robustness.

Relevance Motivates adaptive agent behavior (Normal/Blind/Mute) and unified embedding design.

Paper 2 BLIP – Bootstrapped Vision Language Pretraining Authors Junnan Li, Dongxu Li, Caiming Xiong, Steven CH.

Hoi Reference J.

Li et al, ICML 2022.

Summary BLIP enables high-quality image captioning, vision-language alignment, and contextual representation learning.

Its architecture includes □ A vision encoder for feature extraction □ A transformer decoder for natural-language captioning It outperforms earlier VQA models and supports strong zero-shot generalization.

Relevance Used for generating captions that become semantic anchors for vision–text embedding alignment and memory indexing.

Paper 3 DeepFace – Emotion & Face Understanding Framework Authors Yaniv Taigman, Ming Yang, Marc’Aurelio Ranzato, Lior Wolf Reference Taigman et al, CVPR 2014.

Summary DeepFace introduced deep CNN-based face recognition capable of near-human-level accuracy.

Modern implementations also provide emotion classification output (happy, sad, fear, anger), expanding its utility beyond identity verification.

Relevance Supplies emotional cues from images, which influence memory importance scoring and cognitive reasoning.

Paper 4 Sentence-BERT (SBERT) – Semantic Text Embeddings Authors Nils Reimers, Iryna Gurevych Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 14 Reference Reimers & Gurevych, EMNLP 2019.

Summary SBERT generates dense sentence embeddings optimised for semantic similarity.

Key characteristics □ Siamese/Triplet architecture □ 384–768 dimensional embeddings □ High performance in ranking, clustering, retrieval Relevance Forms the backbone for text queries, caption embeddings, memory comparison, and reasoning.

Paper 5 DistilRoBERTa-Based Emotion Recognition Authors Victor Sanh, Lysandre Debut, Julien Chaumond, Thomas Wolf Reference Sanh et al, 2019.

Summary DistilRoBERTa offers compressed transformer-based models with strong emotional classification performance across categories such as joy, anger, sadness, and neutrality.

Relevance Enables emotional weighting of user text, aiding memory prioritization and reasoning contextualization.

Paper 6 VOSK ASR + Mel-Spectrogram Audio Embeddings Authors AlphaCephei VOSK Development Team Reference VOSK Documentation; Mel-scale foundational work by Stevens, Volkman & Newman (1937).

Summary Audio understanding in AI requires □ Mono/16 kHz normalization □ VAD (Voice Activity Detection) filtering □ Mel-spectrogram conversion □ CNN-based embedding projection VOSK provides reliable offline ASR suitable for assistive and multimodal systems.

Relevance Enables Blind Agent reasoning purely from speech input and supports audio-based emotion and

memory modelling.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 15 Paper 7 Neural Turing Machines & Memory-Augmented Networks Authors Alex Graves, Greg Wayne, Ivo Danihelka Reference Graves et al, NeurIPS 2014.

Summary Neural Turing Machines (NTMs) introduce differentiable read–write memory, enabling neural networks to perform structured reasoning, algorithmic learning, and content-based retrieval.

Relevance Inspirational foundation for the Memory Manager and Memory Retriever that perform multimodal storage, scoring, and Top-N recall.

Paper 8 Gemini Flash 25 – Multimodal Reasoning Architecture Authors Google DeepMind Gemini Research Team Reference Gemini Technical Report, 2024.

Summary Gemini supports advanced multimodal reasoning, causal inference, cross-attention mechanisms, and natural-language explanation synthesis.

Relevance Used as the core reasoning engine for □ Causal inference □ Comparative reasoning □ Human-like explanation generation □ Agent-to-agent modality translation Paper 9 Visual-Linguistic Pretraining for Multimodal Transformers Authors Jingjing Li, Ramprasaath R.

Selvaraju, Arnav Gotmare, Shafiq Joty, Caiming Xiong, Steven CH.

Hoi Reference J.

Li et al, CVPR 2020.

Summary This work introduces unified visual–linguistic pretraining strategies for multimodal transformers.

The model jointly learns vision and language representations through tasks such as masked language modeling, image–text matching, and region-based reasoning.

It demonstrates that pretraining over large-scale datasets significantly improves performance in VQA, captioning, and multimodal retrieval tasks.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 16 Relevance Supports the idea that joint multimodal embeddings improve cross-modal semantic alignment, validating our unified 15D embedding approach for images, audio, and text.

Paper 10 Meta-Learning and Biologically Inspired AI – Towards Adaptive Intelligence Authors Yoshua Bengio, Aaron Courville, Pascal Vincent Reference Y.

Bengio et al, Proceedings of the IEEE, 2019.

Summary This paper explores biologically inspired mechanisms such as synaptic plasticity, meta- learning, and adaptive neural dynamics.

The authors highlight how real brain systems continuously adapt through feedback, reinforcement, and structural reorganization concepts that modern AI systems lack.

The study emphasizes learning-to-learn techniques that enable networks to rapidly adapt to missing or corrupted inputs.

Relevance Forms the theoretical foundation for neuro-adaptive reasoning and agent compensation strategies (Blind and Mute agents), directly supporting the biological inspiration behind your system’s adaptive behavior.

23 Drawbacks of Existing Systems Despite significant progress in multimodal Artificial Intelligence, existing systems still face several limitations that reduce their effectiveness in real-world applications.

Some of the major drawbacks observed in current systems are discussed below.

Dependence on Complete Input Modalities Most existing multimodal systems are designed to operate only when all required input modalities such as image, audio, and text are available.

If one modality is missing, system performance degrades significantly or fails completely.

This limits their applicability in assistive scenarios involving visual or speech impairments.

Limited Adaptive Behavior While humans naturally adapt their reasoning process when a sensory input is unavailable, most AI systems lack such adaptive mechanisms.

Existing models follow fixed processing pipelines and are unable to modify their reasoning strategy based on Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 17 available inputs.

Weak Memory Support Many current AI models process inputs independently without persistent memory.

As a result, they cannot store and reuse multimodal information across interactions.

This leads to loss of context and prevents long-term learning from previous user inputs.

Limited Explanation and Interpretability Most multimodal systems focus on generating outputs or descriptions rather than explaining how conclusions are reached.

They often lack structured reasoning and provide limited insight into the decision-making process, making interpretation difficult.

Absence of Multi-Agent Reasoning Existing systems typically operate as a single unified model.

There is no clear separation of reasoning roles or mechanisms for collaboration between different reasoning strategies.

This limits flexibility in handling different sensory constraints and prevents adaptive interaction across reasoning modules.

Challenges in Cross-Modal Representation Many multimodal models use high-dimensional feature representations that are not well aligned across modalities.

This can result in ☐ Inconsistent semantic interpretation ☐ Reduced accuracy in similarity comparison ☐ Inefficient fusion of image, audio, and text information Such issues affect memory retrieval and overall reasoning consistency.

Sensitivity to Real-World Input Variations Real-world data often contains noise such as unclear images, background audio noise, speech variations, and informal text.

Existing systems may perform poorly under these conditions due to rigid processing and fusion methods.

24 Problem Statement Although multimodal Artificial Intelligence systems have shown improvement in combining image, audio, and text data, most existing systems depend on the availability of all input modalities.

When one or more modalities are missing or degraded, such as in Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 18 visually impaired or speech-impaired scenarios, system performance reduces significantly.

This limitation makes current systems unsuitable for many real-world assistive and interactive applications.

In addition, most existing AI systems follow fixed processing pipelines and do not adapt their reasoning behavior based on available inputs.

Unlike humans, who can rely on alternative senses when one sensory channel is unavailable, current AI models lack adaptive mechanisms inspired by neuroplasticity.

Furthermore, many systems do not maintain persistent multimodal memory, leading to loss of contextual information across interactions.

Existing approaches also typically operate as single-model architectures and do not support specialized reasoning strategies for different sensory conditions.

There is a lack of coordinated multi-agent frameworks where different agents can reason independently based on sensory constraints and share information in suitable formats.

Therefore, the core problem addressed in this project is To design and implement a neuro-adaptive multimodal AI system that can perform effective reasoning when one or more input modalities are unavailable, using a multi-agent architecture and memory-supported reasoning inspired by the concept of neuroplasticity.

The proposed system aims to ☐ Adapt reasoning behavior based on available sensory inputs ☐ Support blind and mute scenarios through specialized agents ☐ Maintain memory to improve contextual consistency ☐ Provide clear explanations suitable for different sensory constraints **25 Proposed Solution** To address the limitations of existing multimodal AI systems, this project proposes a Neuro-Adaptive Multimodal Multi-Agent Reasoning Framework inspired by the concept of neuroplasticity.

In this system, neuroplasticity refers to the ability of artificial agents to adapt their reasoning strategy based on the availability of input modalities, rather than modeling biological neural changes.

The proposed solution is designed to function effectively even when one or more modalities such as image, audio, or text are unavailable.

This makes the system suitable Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 19 for assistive and real-world environments where input conditions are often incomplete or constrained.

The solution is built using a modular architecture that integrates multi-agent reasoning, multimodal embeddings, memory support, adaptive explanation generation, and visualization.

251 Neuro-Adaptive Multi-Agent Architecture The core of the proposed system is a multi-agent framework consisting of three specialized agents.

Each agent is designed to operate under specific sensory conditions and adjust its reasoning process accordingly.

- Normal Agent
 - o Uses all three modalities image, audio, and text
 - o Performs complete multimodal reasoning
 - o Produces both text and audio explanations
- Blind Agent
 - o Operates without image input
 - o Relies on audio and text for reasoning
 - o Generates audio-based explanations suitable for visually impaired users
- Mute Agent
 - o Operates without audio input
 - o Uses image and text information
 - o Produces structured and concise text-based explanations

When a modality is missing, the corresponding agent automatically adapts its reasoning strategy by relying more strongly on the available modalities.

This agent-level adaptation simulates neuroplastic behavior by reorganizing reasoning pathways without retraining models or modifying network weights.

252 Unified Multimodal Embedding Space To enable effective cross-modal reasoning, all input modalities are projected into a shared 15-dimensional embedding space.

This unified representation allows the system to compare, retrieve, and reason across different modalities in a consistent manner.

- Image features are extracted using a ResNet-based encoder and projected into the shared space
- Audio features are obtained using a CNN-based encoder on processed audio signals
- Text inputs are encoded using SBERT-based semantic embeddings and projected

Neuro Adaptive AI Agents Dept. of CS&E, BIT 2025-26 20 into the same space The use of a low-dimensional shared embedding space improves computational efficiency, simplifies memory indexing, and ensures semantic alignment across modalities.

This design enables stable and consistent retrieval of relevant information regardless of input type.

253 Memory-Augmented Reasoning Framework The system includes a cognitive memory module that stores previously processed multimodal information.

- Each memory entry contains
 - Encoded feature vectors
 - Semantic embeddings
 - Emotion-related importance values
 - Contextual summaries

During reasoning, the system retrieves the most relevant memory entries using similarity measures and importance weighting.

Retrieved memories are then combined with current input to enhance reasoning accuracy and contextual consistency.

This memory-supported approach allows the system to maintain continuity across queries and improves explanation quality over time.

254 Reasoning and Explanation Module The proposed system integrates a reasoning engine that generates structured explanations based on

- Current multimodal inputs
- Retrieved memory context
- Agent-specific sensory constraints

The reasoning module produces explanations in formats appropriate to each agent

- Text and audio explanations for the Normal Agent
- Audio-only explanations for the Blind Agent
- Text-only explanations for the Mute Agent

The reasoning process follows a structured pipeline including query understanding, context fusion, reasoning generation, and explanation formatting.

This ensures that outputs are clear, interpretable, and suitable for different users and environments.

255 Cross-Agent Feedback and Translation A unique feature of the system is the agent-to-agent feedback mechanism, which Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 21 enables translation of explanations across modalities

- Audio explanations generated by the Blind Agent can be converted into concise text summaries for the Mute Agent
- Text explanations from the Mute Agent can be transformed into natural audio explanations for the Blind Agent

This cross-agent interaction allows collaborative reasoning and improves accessibility by ensuring that information can be shared in formats suitable for different sensory conditions.

256 Visualization and Deployment Framework To support evaluation and demonstration, the system includes visualization and deployment components.

- Multimodal Input Visualization
 - o Uses PCA-based 3D visualization to represent image, audio, and text embeddings
 - o Helps analyze semantic alignment across modalities
- Reasoning Space Visualization
 - o Compares reasoning outputs from Normal, Blind, and Mute agents
 - o Demonstrates adaptive behavior under sensory constraints
- Web-Based Deployment
 - o Implemented using the Flask framework
 - o Supports multimodal input upload, agent selection, reasoning display, audio playback, and visualization
 - o Provides an interactive interface for system evaluation

CHAPTER 3 REQUIREMENT ENGINEERING Dept. of CS&E, BIT 2025-26 22 CHAPTER 3 REQUIREMENT ENGINEERING Requirement Engineering is the process of identifying and defining the needs of the system before development begins.

This chapter describes the requirements necessary to design and implement the proposed Neuro-Adaptive Multi-Agent System.

It explains the hardware and software requirements, functional and non-functional requirements, and the overall constraints of the system.

31 Software and Hardware Tools Used
311 Software Tools The system extensively uses modern AI development frameworks and multimodal processing libraries
Category Tool / Technology Description
Operating System Windows 10 / Ubuntu Linux
Development & execution environment supporting Python ML stack, CUDA acceleration, and Flask deployment.

Programming Language Python 3.10+
Core language for multimodal processing, agent reasoning, encoders, Gemini Flash API, & Flask backend.

Development Environment VS Code, Jupyter Notebook
Used for iterative model development, debugging, dataset handling, and embedding visualization.

Backend Framework Flask
Web service layer enabling file uploads, agent execution, visualization endpoints, and API responses.

Deep Learning Framework PyTorch
Used to train image, audio, and text encoders; supports 15D latent embedding generation and model inference.

Transformer Framework Hugging Face Transformers
Provides SBERT, DistilRoBERTa emotion classifier, and tokenizer utilities for text processing.

Sentence Embedding Model Sentence-BERT (all-MiniLM-L6-v2)
Generates 384-dimensional text embeddings for reasoning & memory retrieval.

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23 Vision Models ResNet-18, BLIP ResNet-18
used for image feature extraction; BLIP for automated image captioning.

Face & Emotion Analysis DeepFace
Used for facial emotion recognition in images (dominant emotion extraction).

Audio Processing Libraries Librosa, PyDub, WebRTC-VAD, Wave
Used for audio resampling, mono conversion, noise filtering, VAD speech detection, and Mel-spectrogram features.

Speech-to-Text Engine VOSK
Offline ASR Converts audio to text without internet; robust in noisy environments.

Multimodal Fusion Tools NumPy, SciPy, scikit-learn (PCA)
Used for embedding fusion, similarity computations, and 3D PCA visualization of semantic spaces.

Knowledge & Reasoning Engine Google Gemini Flash 2.5 API
Performs causal reasoning, comparative analysis, human-like explanation, and blind/mute feedback synthesis.

Frontend Technologies HTML5, CSS3, Bootstrap 5, JavaScript
GUI for agent control, visualization, and dynamic multimodal input preview.

Visualization Libraries matplotlib, mpl_toolkits (3D)
Generates 3D modality-space plots, reasoning-space PCA graphs, and visual analytics.

Serialization Tools JSON
Stores memory embeddings, knowledge base items, and structured reasoning outputs.

Table 31 Software tools
312 Hardware Tools The system requires a mid-range computing environment equipped with a multi-core CPU, optional CUDA-enabled GPU, 8–16 GB RAM, sufficient SSD storage, and standard audio–visual peripherals to support efficient multimodal processing, real-time inference, and neuro-adaptive agent interaction.

Hardware Component Specification / Model Purpose in Project
Processor (CPU) Intel Core i5/i7 (8th Executes multimodal processing
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24 Gen or above) AMD Ryzen 5/7 (3000 series or above)
pipelines (image/audio/text), VOSK ASR, SBERT inference, Flask backend, and memory retrieval operations.

Graphics Processor (GPU) Optional but recommended NVIDIA GTX 1650 / RTX 2060 / RTX 3060 or higher with CUDA support
Accelerates PyTorch-based encoders (ResNet18, Audio CNN, SBERT projection layers), BLIP captioning, DeepFace emotion analysis, and 3D PCA visualization.

Memory (RAM) 8 GB minimum 16 GB recommended
Supports in-memory operations for embeddings (384D → 15D), JSON memory store, multimodal fusion buffers, and transformer inference without latency.

Storage Minimum 10 GB free storage (SSD preferred)
Used for storing pre-trained models (BLIP, SBERT,

Vosk), memory_store JSON, embeddings, datasets, and Flask server assets.

Microphone Any 16 kHz mono- compatible microphone Required for speech input, VAD detection, and blind-agent audio reasoning.

Audio Output Device Speakers / Headphones Used by Blind-Agent for generated audio explanations via pyttsx3 TTS.

Camera (Optional) USB webcam / system- integrated camera Supports real-time image input for captioning and emotion analysis in future extension.

GPU Drivers NVIDIA CUDA Toolkit + cuDNN Enables optimized PyTorch model execution and multimodal encoder acceleration.

Network Connection Stable internet connection (for Gemini Flash API) Required for causal reasoning, comparative insights, and agent collaboration responses.

Table 32 Hardware tools Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 25 32 Conceptual/Analysis Modeling Conceptual and analysis modeling is used to understand the overall structure and behavior of the proposed Neuro-Adaptive Multi-Agent System before actual implementation.

It provides a clear view of how users interact with the system, how different agents operate, and how data flows between system components.

This modeling helps represent the internal working of the system using diagrams such as use case diagrams, activity diagrams, and sequence diagrams.

It explains how multimodal inputs (image, audio, and text) are processed, how agents respond under different sensory conditions, and how memory and reasoning modules support decision-making.

These models ensure clarity in system behavior, validate functional requirements, and serve as a blueprint for system design and development across the Normal, Blind, and Mute agents.

321 Use Case Diagram The Use Case Diagram illustrates the functional interaction between external entities and the proposed Neuro-Adaptive Multi-Agent System.

It presents a high-level view of how the system operates from the user's perspective and how internal services such as memory management support the reasoning process.

Actors The diagram consists of two primary actors □ User The user interacts with the system by providing multimodal inputs such as text, audio, or images and receives the reasoning output generated by the system.

□ Memory Manager The Memory Manager is responsible for storing previously generated reasoning outputs and retrieving relevant information to support adaptive reasoning.

Use Case Flow The system workflow begins when the user provides multimodal input.

The system then performs preprocessing to clean and normalize the inputs, followed by feature extraction from each available modality.

These extracted features are fused to form a unified representation.

Based on the availability of input modalities, the system activates the appropriate agent (Normal, Blind, or Mute).

The selected agent performs adaptive reasoning and generates an output in text or audio format.

During this process, the Memory Manager retrieves Neuro Adaptive AI Agents Dept.

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Finally, the generated response is stored back into memory for future use.

Figure 31 Use Case Diagram □ The interaction begins when the user provides multimodal input, which may include text, audio, and image data, thereby initiating the system's processing pipeline.

□ Each input modality undergoes preprocessing to ensure data consistency and quality.

This includes audio cleaning and normalization, image resizing and normalization, and text formatting.

□ The system then performs feature extraction using modality-specific encoders such as the Image Encoder, Audio Encoder, and Text Encoder.

□ The extracted feature embeddings are fused into a unified multimodal representation, enabling consistent and comparable reasoning across different input types.

□ Based on the availability of input modalities, the system selects and activates the appropriate agent, namely the Normal Agent, Blind Agent, or Mute Agent.

□ The selected agent performs adaptive reasoning and generates a suitable output in either textual or audio form.

□ During the reasoning process, the Memory Manager retrieves relevant past memories to enhance contextual understanding and refine the response.

□ After generating the final output, the system stores the newly generated reasoning instance in memory for future reference and learning.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 27 □ Finally, the refined response is presented to the user, completing the interaction cycle.

322 Sequence Diagram A Sequence Diagram represents the time-ordered interaction between different components of the Neuro-Adaptive Multi-Agent System.

It shows how messages are exchanged among system modules during the execution of a user request.

In this system, the sequence begins when the user submits multimodal inputs, such as image, audio, or text, to the system interface.

These inputs are forwarded to the Agent Manager, which coordinates the overall processing workflow.

The Agent Manager initiates preprocessing and embedding generation by invoking the respective encoders for image, audio, and text.

Based on the availability of input modalities, it then activates the appropriate agent Normal Agent, Blind Agent, or Mute Agent.

Once the agent is activated, the system performs memory retrieval to obtain relevant past reasoning instances that support contextual understanding.

The agent then executes adaptive reasoning, and when required, invokes the Gemini-based reasoning module to generate structured and meaningful explanations.

After reasoning is completed, the generated explanation is returned to the Agent Manager, which forwards the final response to the user.

This sequence ensures coordinated interaction between perception modules, agents, memory components, and the reasoning engine.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 28 Figure 32 Sequence Diagram 323 Activity Diagram An Activity Diagram represents the overall workflow of the system, highlighting the sequence of operations, decision points, and data flow involved in processing user requests.

It focuses on how different activities are performed from input acquisition to final response generation.

The workflow begins when the user provides input to the system.

The input may include one or more modalities such as image, audio, or text.

Once the input is received, the system performs preprocessing and embedding generation, where each available modality is converted into its corresponding feature representation.

After embedding generation, the system checks the availability of input modalities.

Based on this condition, the appropriate agent is selected □ The Normal Agent is activated when all modalities are available.

□ The Blind Agent is activated when image input is missing.

□ The Mute Agent is activated when audio input is missingNeuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 29 If a modality is unavailable, the system relies on stored memory and contextual information to support reasoning.

The active agent then retrieves relevant past memory entries and combines them with the current input to enhance contextual understanding.

The refined information is passed to the reasoning module, which generates an explanation suited to the selected agent.

The system then presents the output to the user in an appropriate format such as text or audio.

Visualization components may also be used to display embedding or reasoning information when required.

Finally, the generated response and interaction details are stored in memory for future reference.

This workflow repeats for subsequent user inputs until the interaction is completed.

Figure 33 Activity Diagram Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 30 33 Software Requirements Specification (SRS) The Software Requirements

Specification (SRS) defines the complete set of functional and non-functional requirements required for the

successful development and operation of the proposed Neuro-Adaptive Multi-Agent System.

This document translates user expectations into well-defined system capabilities and serves as a foundation for system design, implementation, and validation.

The SRS ensures that the system behaves consistently, reliably, and efficiently across different usage scenarios and sensory constraints.

331 User Requirements User requirements specify what the system should provide from an end-user perspective. The proposed system is intended to be accessed through a web-based interface that supports multimodal interaction.

Users must be able to upload text, image, and audio inputs individually or in combination, depending on their needs and available data.

The system should allow users to explicitly select the desired agent type Normal, Blind, or Mute so that reasoning is performed according to the chosen sensory configuration.

The system must generate context-aware reasoning outputs that adapt to the selected agent.

For the Normal Agent, the user should receive comprehensive multimodal explanations.

For the Blind Agent, the system should primarily deliver audio-based explanations, while for the Mute Agent, the system should provide clear and structured textual explanations.

Users should also be able to view visual representations of multimodal embeddings and reasoning outputs using interactive 3D PCA plots, which help in understanding how different modalities influence reasoning.

In addition, the system should allow users to preview uploaded images and audio files before execution to ensure correctness of inputs.

The user should receive refined explanations that incorporate relevant past interactions stored in memory, enabling continuity across sessions.

Importantly, the system must remain functional even when one or more input modalities are missing, ensuring usability in real-world scenarios such as visual or speech impairment.

3311 Non-Functional User Requirements From a usability standpoint, the system should offer a simple, intuitive, and user-friendly interface that can be operated without prior technical knowledge.

The response time of Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 31 the system should be reasonably fast, allowing users to receive outputs within a few seconds after submission.

The system must be robust and should not crash or produce undefined behavior when noisy, incomplete, or invalid inputs are provided.

Furthermore, user-uploaded data must be handled securely, ensuring that sensitive information is not exposed or misused.

332 System Requirements System requirements define the internal functionalities and technical capabilities necessary to satisfy user requirements.

The system must support independent preprocessing pipelines for image, audio, and text inputs.

Images should be normalized and processed using a dedicated image encoder, audio inputs should undergo preprocessing such as resampling and speech extraction before feature encoding, and textual inputs should be cleaned and encoded using a sentence-level transformer model.

The system must extract embeddings from each modality using specialized encoders, including a ResNet-based image encoder, a CNN-based audio encoder, and an SBERT-based text encoder.

These embeddings must be projected into a unified 15-dimensional latent space to ensure consistent cross-modal representation.

This shared embedding space enables effective comparison, retrieval, and reasoning across heterogeneous modalities.

Based on the availability of input modalities, the system must dynamically activate the appropriate reasoning agent.

When all modalities are present, the Normal Agent should be selected.

If image input is unavailable, the Blind Agent must operate using audio and text data.

If audio input is missing, the Mute Agent should rely on image and text information.

This adaptive agent selection mechanism allows the system to simulate sensory compensation and maintain reasoning functionality under constrained conditions.

The system must include a memory management module that stores multimodal experiences along with semantic relevance and contextual metadata.

This memory should be queried during reasoning to retrieve the most relevant past interactions, thereby enhancing contextual understanding and explanation quality.

Newly generated reasoning outputs must be stored back into memory to support future interactions.

The reasoning module should generate structured, context-aware explanations using an Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 32 integrated reasoning engine.

The system must support both text generation and audio synthesis depending on the active agent.

Additionally, the system should provide visualization modules capable of generating 3D PCA plots for multimodal embeddings and agent-wise reasoning comparisons.

3321 Non-Functional System Requirements In terms of performance, the system should efficiently handle multimodal data processing and reasoning, leveraging available computational resources such as GPU acceleration when present.

The system must ensure low-latency inference to maintain a smooth user experience.

From a reliability perspective, the system should gracefully handle incomplete or corrupted inputs and ensure that memory retrieval always produces a valid ranked output.

Security is a critical requirement; uploaded files must be stored in isolated and secure directories, and unauthorized access to memory or system logs must be prevented.

The system should be compatible with commonly used web browsers such as Chrome, Firefox, and Edge, and should operate on standard platforms including Windows and Linux using Python-based environments.

Finally, the system should follow a modular and maintainable architecture, allowing individual components such as encoders, memory modules, or reasoning engines to be updated or replaced without affecting the overall system structure.

CHAPTER 4 PROJECT PLANNING Dept.

of CS&E, BIT 2025-26 33 CHAPTER 4 PROJECT PLANNING Project planning establishes a systematic and organized roadmap for the successful development of the Neuro-Adaptive Multi-Agent System.

It ensures that the project progresses in a controlled manner by defining clear phases, allocating resources efficiently, managing risks, and monitoring milestones throughout the development lifecycle.

Effective planning is essential to transform conceptual ideas into a fully functional system within the given academic timeframe.

41 Project Planning and Scheduling The project is executed in multiple phases based on Incremental and Iterative Development, ensuring continuous testing, improvement, and integration of multimodal agents and memory systems.

Phase Duration Major Activities / Tasks Phase 1 Requirement Analysis Week 1– 2 □ Identify project scope and objectives □ Analyze multimodal processing needs (text, audio, image) □ Define system architecture & agent roles (Normal, Blind, Mute) □ Prepare Software Requirement Specification (SRS) Phase 2 System Design Week 3– 4 □ Design Image, Audio, and Text encoder modules □ Define unified 15D embedding framework □ Design Memory Manager & Retrieval Engine □ Construct reasoning workflow using Gemini Flash □ Prepare UML diagrams (Use Case, Sequence, Activity) Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 34 Phase 3 Data Processing & Model Integration Week 5– 7 □ Implement preprocessing pipelines for all modalities □ Integrate BLIP, DeepFace, SBERT, VOSK □ Develop custom encoders for multimodal embeddings □ Validate embedding quality and performance Phase 4 Multi-Agent Reasoning System Week 8– 10 □ Implement Normal, Blind, and Mute agents □ Add neuro-adaptive logic for missing modalities □ Integrate memory-aware reasoning pipeline □ Implement agent-to-agent feedback (Mute ↔ Blind) Phase 5 Backend Development Week 11–12 □ Build Flask backend server □ Implement routes for agent execution □ Provide API for 3D PCA visualization □ Implement secure file upload & media handling Phase 6 Frontend Development Week 13–14 □ Design multimodal input UI (text/audio/image) □ Build agent execution dashboard □ Add result preview modules (audio player, text display) □ Integrate reasoning visualization screens Phase 7 Testing & Performance Evaluation Week 15–16 □ Perform unit testing for all modules □ Conduct integration testing of multi-agent system □ Evaluate reasoning accuracy and Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 35 consistency □ Benchmark memory retrieval and optimize latency Phase 8

Documentation & Final Report Week 17–18 □ Prepare complete project documentation □ Document architecture, APIs, and algorithms □ Compile test results, graphs, and snapshots □ Prepare presentation slides and demonstration material Table 41 Project Development Phases Figure 41 Gantt Chart 411 Team Role Allocation Team Member Role Responsibilities Shashi Madari Lead Full-Stack Developer & Reasoning Engine □ Integrate Gemini Flash 25 for causal reasoning & explanation generation □ Develop Flask backend APIs, routing, and agent execution engine Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 36 □ Implement 3D visualization modules (PCA reasoning + modality plots) □ Build and refine frontend UI/UX for multi-agent interaction Rohit K H AI/ML Engineer (Vision & Audio Models) □ Implement BLIP, DeepFace, audio preprocessing, VOSK speech-to-text □ Build and train custom Image/Audio encoder models □ Optimize multimodal embedding generation (15D latent space) □ Evaluate model performance and feature extraction pipelines Srivallabha K M System Architect & Developer □ Define overall system architecture (encoders, memory system, agents) □ Oversee integration of multimodal pipelines (image, audio, text) □ Ensure system-level coordination between backend, frontend, and reasoning modules □ Guide code structure, optimization, and deployment strategy Abhishek Srivatsav Memory Manager & Testing Lead □ Develop Memory Manager and Retrieval Engine (Top-N recall, similarity matching) □ Ensure memory update logic for adaptive learning cycles □ Perform unit testing, integration testing, and system verification □ Handle performance benchmarking and error resolution Table 42 Team Role Allocation CHAPTER 5 SYSTEM DESIGNDept.

of CS&E, BIT 2025-26 37 CHAPTER 5 SYSTEM DESIGN 51 System Architecture The overall system architecture integrates multimodal perception, adaptive agent selection, memory-assisted reasoning, and explanation generation into a unified processing pipeline.

Raw sensory inputs in the form of image, audio, and text are first processed through dedicated modality-specific encoders, which convert heterogeneous data into structured feature representations.

These representations are projected into a shared low-dimensional latent embedding space to ensure semantic consistency and enable cross-modal comparison.

511 Architectural Overview The proposed system is designed using a layered architecture, where each layer performs a specific function in transforming raw multimodal inputs into meaningful and adaptive reasoning outputs.

This structured design improves modularity, interpretability, and robustness while enabling neuro-adaptive behavior when certain sensory modalities are missing or degraded.

Figure 51 Block Diagram Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 38 Input Layer (Raw Modalities) The Input Layer is responsible for receiving multimodal data from the user.

Each modality is accepted independently, allowing flexible interaction and ensuring system robustness under incomplete input conditions.

□ Image Input Captures visual information such as objects, scenes, diagrams, facial expressions, or environmental cues.

Visual input supports spatial understanding and contextual interpretation.

□ Audio Input Represents spoken language, environmental sounds, or other auditory signals.

Audio input contributes linguistic content and contextual cues after speech processing.

□ Text Input Provides structured language-based descriptions, questions, or captions directly supplied by the user.

Text input forms the semantic backbone for reasoning across all agents.

The independence of inputs allows the system to operate effectively even when one or more modalities are missing, noisy, or inconsistent.

This flexibility is essential for supporting neuro-adaptive behavior.

Modality Encoding Layer Each modality is processed using a dedicated encoder optimized for its data type.

These encoders transform heterogeneous raw inputs into numerical embeddings with consistent dimensionality.

□ ResNet-18 Image Encoder → 256-Dimensional Embedding Extracts visual features such as object patterns, spatial relationships, and global contextual cues from image input.

□ Sentence-BERT Text Encoder → 256-Dimensional Embedding Converts textual input into dense semantic embeddings that capture meaning, intent, and contextual relationships.

□ CNN-Based Audio Encoder → 256-Dimensional Embedding Processes audio signals to learn spectro-temporal patterns, speech characteristics, and semantic cues.

By converting all modalities into homogeneous vector representations, this layer enables effective cross-modal alignment and comparison.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 39 Figure 52 Input handling and training loop architecture Shared Neuro-Adaptive Latent Space (15D) The embeddings generated from each modality are projected into a shared 15- dimensional latent space using trainable projection layers.

This compact representation serves as a common reasoning space across all modalities.

The shared latent space enables the system to □ Align different modalities into a common conceptual representation □ Perform cross-modal reasoning □ Execute efficient similarity-based retrieval □ Support modality compensation □ Enable agent-level adaptation using unified embeddings This 15-dimensional space functions as the system's core reasoning layer, where semantic relationships are formed and compared efficiently.

Modality Compensation Engine When one or more modalities are missing or corrupted, the system activates the Modality Compensation Engine.

This component reconstructs or approximates the missing Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 40 representation using correlations learned from available modalities within the shared latent space.

This mechanism ensures that □ The latent representation remains complete and meaningful □ The system does not fail due to missing input □ Agents receive consistent embeddings for reasoning This behavior mirrors biological neuroplasticity, where weakened sensory pathways are compensated by strengthening remaining modalities.

Multi-Agent Reasoning Layer Three agents reason over the projected embeddings, each specializing in different sensory availability conditions and performing neuro-adaptive reasoning depending on which modalities are present or compensated.

Each agent follows a distinct computational pathway and internally constructs its own reasoning hypothesis □

Normal Agent (Image + Audio + Text) Operates when all modalities are available and produces the most comprehensive reasoning output by fusing all modality embeddings.

□ Blind Agent – Uses Audio + Text Functions without visual input and relies on linguistic and auditory semantics to generate reasoning, simulating visually impaired cognition.

□ Mute Agent – Uses Image + Text Operates without audio input and emphasizes visual–textual reasoning to compensate for missing speech information.

Each agent follows a distinct computational pathway.

The Agent Manager evaluates modality quality, confidence levels, and semantic consistency to select the most appropriate reasoning behavior or combine agent outputs when necessary.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 41 Memory Manager & Retrieval Layer The memory module acts as long-term semantic storage, holding multimodal knowledge with compact latent keys and richer semantic values.

Each memory entry consists of □ Keys 15D latent vectors used for similarity indexing □ Values 384D semantic embeddings used for richer contextual recall Memory retrieval is performed using a combined similarity score derived from □ similarity in semantic space □ similarity in latent reasoning space □ additional boosting factors Where o similarity in 384D semantic representation o similarity in 15D reasoning space o boosting term defined as This mechanism enables efficient and contextually relevant recall of past experiences, strengthening reasoning even when current inputs are incomplete.

Reasoning Fusion Engine The Reasoning Fusion Engine integrates □ outputs from multiple agents □ retrieved memory context □ cross-modal semantic representations It resolves conflicting interpretations, reinforces consistent evidence, and generates a unified reasoning vector.

This stage acts as a decision-making core that consolidates multisensory evidence into coherent reasoning.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 42 Figure 53 Reasoning Stage architecture Output Layer The final reasoning stage produces the system's external response by converting fused multimodal understanding into human-interpretable

explanations.

This layer receives the unified reasoning vector from the fusion engine, incorporates confidence information from all contributing agents, and formats the output into a structured explanation.

The Output Layer generates □ Explanation A natural-language statement describing the inferred reasoning and interpreted meaning of the multimodal input.

□ Interpretation The contextual understanding refined by agent decisions, modality focus, and memory-based recall.

□ Final answer A conclusive response synthesized after evaluating all available sensory evidence.

□ Cross-agent feedback A feedback signal returned to the Agent Manager that reflects reasoning confidence, precision, and success, which helps dynamically Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 43 adjust future agent selection and modality prioritization.

The Output Layer therefore completes the reasoning pipeline by providing a coherent final answer along with interpretability, confidence scoring, and adaptivity, ensuring neuro-inspired continuous improvement across multiple reasoning cycles.

Figure 54 Output Stage architecture 52 Component Design / Module Decomposition The proposed Neuro-Adaptive Multi-Agent System is decomposed into a set of well-defined functional modules, each responsible for a specific stage of multimodal data processing and reasoning.

This modular decomposition simplifies system development, improves clarity, enables parallel processing, and allows individual components to be enhanced or replaced without affecting the overall architecture.

The modular design supports adaptive multimodal reasoning by clearly separating perception, representation, memory, agent-based reasoning, and output generation, while maintaining a unified cognitive inference pipeline.

521 Module 1 – Input Processing Module The Input Processing Module prepares raw multimodal inputs for neural processing by converting them into standardized and machine-readable formats.

□ Image preprocessing Images are resized, normalized, and converted into tensors suitable for deep neural networks, ensuring consistent spatial and pixel-level representation.

□ Audio preprocessing Audio signals are converted to mono format, resampled to Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 44 a standard frequency, and filtered using Voice Activity Detection (VAD).

Speech segments are identified and prepared for feature extraction.

□ Text preprocessing Input text is cleaned, normalized, and formatted for semantic encoding, ensuring compatibility with transformer-based language models.

□ Modality availability detection The module identifies missing or invalid inputs and signals downstream components to adjust processing accordingly.

This module ensures that all subsequent components receive structured, normalized, and reliable inputs, regardless of modality availability.

522 Module 2 – Feature Encoder Module This module contains modality-specific encoders that extract high-level semantic features from processed inputs.

The encoders used are □ Image Encoder A ResNet-18-based encoder extracts visual features such as object patterns, spatial structure, and scene context.

□ Audio Encoder A CNN-based model processes audio features derived from speech signals to capture temporal and acoustic patterns.

□ Text Encoder A Sentence-BERT (SBERT) model generates dense semantic embeddings representing sentence meaning and contextual relationships.

Each encoder produces a fixed-length feature vector, which provides a modality-specific semantic representation for further alignment.

523 Module 3 – Latent Projection and Modality Compensation Module To align heterogeneous modality embeddings into a unified reasoning space and handle missing sensory information.

□ Latent space projection Encoder outputs are projected into a shared 15-dimensional latent embedding space, enabling uniform comparison across modalities.

□ Modality reliability assessment The system evaluates the availability and quality of each modality.

□ Modality compensation When a modality is missing, the system relies on available modality embeddings and stored memory context to preserve reasoning continuity.

This module ensures that reasoning remains stable and semantically meaningful even under incomplete or degraded input conditions.

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of CS&E, BIT 2025-26 45 524 Module 4 – Multi-Agent Reasoning Module This module implements a multi-agent reasoning framework, where each agent operates under different sensory constraints.

The system includes

- Normal Agent Uses image, audio, and text for full multimodal reasoning.

- Blind Agent Operates without image input and performs reasoning using audio and text.

- Mute Agent Operates without audio input and relies on image and text.

Each agent generates a reasoning representation based on its available modalities.

An Agent Manager selects or prioritizes agent outputs based on modality availability and confidence, introducing adaptive behavior similar to human sensory compensation.

525 Module 5 – Memory System Module The Memory System provides persistent storage and retrieval of multimodal experiences, enabling context-aware reasoning.

- Memory storage Stores multimodal reasoning instances using compact latent representations.

- Similarity computation Matches new queries with stored memories using latent and semantic similarity measures.

- Contextual boosting Retrieval is enhanced using modality relevance, keyword overlap, and importance weighting.

Each memory entry is structured as

- Key (15D) Latent representation for indexing
- Value (semantic embedding) Rich contextual information
- Metadata Modality type, importance, and emotional cues
- Timestamp Maintains temporal relevance

This module allows the system to reuse prior knowledge and improve reasoning consistency over time.

526 Module 6 – Reasoning Fusion and Output Module The final module integrates information from

- active agent reasoning outputs,
- retrieved memory context,
- and semantic representations to generate a unified and coherent explanation.

- A final context-aware explanation

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- Modality-adaptive response format (text or audio)
- Confidence and feedback signals for future agent selection

This module ensures that the system produces interpretable, reliable, and adaptive responses, completing the multimodal reasoning pipeline.

53 Interface Design This section defines how the individual modules communicate with one another using well-structured, standardized internal interfaces.

Proper interface design ensures smooth data flow between modules, reduces integration complexity, and enables modularity and scalability of the entire multimodal reasoning pipeline.

Each interface defines input formats, output formats, and internal processing agreements such that modules remain interoperable regardless of their internal implementation.

Interface design defines how individual modules of the Neuro-Adaptive Multi-Agent System interact and exchange data.

A clear interface structure ensures smooth data flow between components, reduces coupling, and enables independent modification of modules without affecting the overall system.

The system follows a modular, service- oriented interaction style where each module exposes clearly defined inputs and outputs.

531 Input Processing Interface The Input Processing Interface acts as the entry point for all user-provided data through the Flask web interface.

Inputs

- Image file (optional)
- Audio file (optional)
- Text input (optional)

Outputs

- Preprocessed image tensor (if image exists)
- Preprocessed audio tensor / transcript (if audio exists)
- Cleaned text string (if text exists)
- Modality availability flags

This interface ensures that each modality is independently validated and forwarded only if available.

Missing modalities are explicitly identified and communicated to downstream components

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of CS&E, BIT 2025-26 47 532 Encoder Interface The Encoder Interface defines how preprocessed inputs are converted into numerical feature representations.

Each modality is processed independently through its respective encoder

- Image Encoder Interface
- Input

Preprocessed image tensor o Output Fixed-length visual embedding □ Audio Encoder Interface o Input Preprocessed audio representation o Output Fixed-length audio embedding □ Text Encoder Interface o Input Raw or summarized text string o Output Fixed-length text embedding All encoder outputs are normalized vectors and are forwarded to the latent alignment stage.

This interface ensures consistent semantic representation regardless of modality type.

533 Latent Representation Interface This interface defines the transformation of encoder outputs into a shared reasoning space.

Inputs □ Image embedding (if available) □ Audio embedding (if available) □ Text embedding (if available)

Outputs □ Unified low-dimensional latent embeddings □ Modality-aware embedding list The latent representation interface enables consistent similarity comparison and reasoning across different modalities.

Missing embeddings are simply excluded rather than causing pipeline failure.

534 Agent Interface The Agent Interface provides a unified mechanism for invoking agent-specific reasoning logic.

Each agent follows the same callable structure Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 48 Agentrun(query_embeddings, memory_context) Inputs □ Available latent embeddings □ Retrieved memory context □ Agent type (Normal / Blind / Mute) Outputs □ Reasoned sentence □ Intermediate reasoning metadata □ Confidence indicators Internally, agents differ in modality usage, but externally they expose a consistent interface.

This abstraction allows seamless switching between agents based on modality availability.

535 Memory Interface The Memory Interface defines interaction with the persistent cognitive memory system.

Memory Storage Interface □ Input New reasoning instance with metadata □ Action Append to persistent memory store Memory Retrieval Interface □ Input Query embeddings □ Output Ranked list of relevant memory entries This interface hides retrieval complexity (similarity calculation, ranking, weighting) and allows memory logic to evolve independently of agent or reasoning logic.

536 Reasoning Engine Interface The Reasoning Engine Interface connects the system to the Gemini Flash reasoning model.

Inputs □ User query □ Retrieved memory context □ Agent-specific reasoning data Outputs □ Human-readable explanation (text) □ Optional audio explanation (for blind agent) □ Summarized reasoning output Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 49 This interface ensures that reasoning generation remains decoupled from perception and memory components, while still being context-aware.

537 Output Interface The Output Interface delivers the final system response to the user via the web interface.

Outputs include □ Final explanation (text and/or audio) □ Agent type used □ Supporting facts and knowledge □ Visualization links (if generated) □ Execution logs This interface supports interpretability, transparency, and adaptive feedback, completing the neuro-adaptive reasoning loop.

537 User Interface Design The User Interface (UI) of the Neuro-Adaptive Multi-Agent System is designed as a web- based interactive platform using HTML, CSS, JavaScript, Bootstrap, and Flask.

The interface enables seamless interaction between the user and the underlying multimodal reasoning architecture while maintaining clarity, accessibility, and responsiveness.

The UI acts as a unified interaction layer through which users can provide multimodal inputs, select agent types, visualize internal representations, and receive adaptive reasoning outputs.

Landing Page Interface The landing page serves as the entry point to the system.

It introduces the concept of neuro-adaptive intelligence and briefly explains the role of Normal, Blind, and Mute agents.

A prominent action button allows users to launch the main system interface.

The design emphasizes clarity, minimalism, and thematic consistency with cognitive intelligence concepts.

Landing Page

of CS&E, BIT 2025-26 50 inputs.

Launch Multi-Agent System Main Interaction Interface The main interface allows users to provide multimodal inputs and initiate agent-based

reasoning.

The layout is divided into clearly separated sections for input acquisition.

Text Input Section Users can enter natural language queries that guide the reasoning process.

`<textarea name=text_input rows=4 placeholder=Enter your question or statement></textarea>` Image Upload Section Users can upload an image file, which is immediately previewed to confirm correctness.

`<input type=file id=image_file name=image_file accept=image/* onchange=previewImage(event)>` function `previewImage(event) { const reader = new FileReader(); reader.onload = e => { document.getElementById(imagePreview).innerHTML = ``; }; reader.readAsDataURL(event.target.files[0]); }` Audio Upload Section Users can upload speech or audio files, with an embedded audio player for preview.

`<input type=file id=audio_file name=audio_file accept=audio/* onchange=previewAudio(event)>` function `previewAudio(event) {` Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 51 `const reader = new FileReader(); reader.onload = e => { document.getElementById(audioPreview).innerHTML = `<audio controls src=${e.target.result}></audio>`; }; reader.readAsDataURL(event.target.files[0]); }` Each modality is optional, allowing the system to demonstrate neuro-adaptive reasoning when certain inputs are missing.

Agent Execution Interface The interface provides separate execution panels for each agent, allowing direct comparison of reasoning behavior.

`<div class=agent-card> <div class=agent-header>  NORMAL AGENT</div> <button onclick=runAgent(normal)>Run Normal Agent</button> <div id=normalOutput></div> </div> <div class=agent-card> <div class=agent-header>   BLIND AGENT</div> <button onclick=runAgent(blind)>Run Blind Agent</button> <div id=blindOutput></div> </div> <div class=agent-card> <div class=agent-header>  MUTE AGENT</div> <button onclick=runAgent(mute)>Run Mute Agent</button> <div id=muteOutput></div> </div>` Each agent panel displays ☐ the reasoning sentence, ☐ text explanations (Normal and Mute agents), ☐ audio explanations (Blind agent), ☐ cross-agent feedback where applicable.

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of CS&E, BIT 2025-26 52 Visualization Controls The UI supports two distinct visualization features that enhance interpretability.

3D Input Modality Visualization This feature visualizes the relative positions of image, audio, and text embeddings in a 3D semantic space.

`<button id=generate3DButton>  Generate 3D Input Visualization</button>` `fetch(/generate_3d, { method POST, body formData })then(res => res.json()) then(data => window.open(data.redirect_url));` 3D Reasoning Space Visualization Once all agents complete reasoning, the user can visualize how agent reasoning differs in embedding space.

`<button id=visualizeReasoningButton>  Visualize Reasoning Space</button>` This visualization highlights neuro-adaptive shifts between Normal, Blind, and Mute agents.

54 Data Structure Design Data Structure Design defines how information is represented, stored, exchanged, and processed internally within the Neuro-Adaptive Multi-Agent System.

Since the system is implemented using Python, Flask, and deep learning frameworks, dynamic data structures such as dictionaries, lists, JSON objects, and tensors are used instead of rigid static structures.

These data structures ensure flexibility, scalability, and efficient handling of multimodal data.

Well-defined data representations enable consistent feature alignment, memory retrieval, agent reasoning, and explanation generation across the entire multimodal pipeline.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 53 541 Multimodal Query Data Structure Each user interaction is internally represented as a query dictionary, where every available modality contributes a structured entry.

A single query may consist of one or more modality-specific entries stored in a list.

Each modality entry follows the structure ☐ type Specifies the modality (image, audio, or text) ☐ query_text Human-readable textual description (caption, transcript, or summary) ☐ key Low-dimensional latent embedding (15D) used for similarity matching ☐ value High-dimensional semantic embedding (Sentence-BERT output) ☐

meta Metadata such as emotion, importance score, and modality-specific attributes Example Image Query Entry { type image, query_text A diagram showing Einsteins mass-energy equivalence, key [012, -008, 034, , 005], value [0021, -0117, 0309, , -0042], meta { emotion neutral, importance 06, caption A diagram illustrating $E = mc^2$ } } Example Audio Query Entry { type audio, query_text Emotion confident, Text energy equals mass times speed of light squared, key [-022, 019, 007, , -011], value [0184, -0091, 0266, , 0033], meta { emotion confident, importance 10, transcribed energy equals mass times speed of light squared } } This unified structure allows the retrieval and reasoning modules to process all modalities uniformly.

542 Encoder Output Representation Encoder outputs are handled as PyTorch tensors or NumPy arrays, depending on processing stage.

Each encoder produces a normalized feature vector which is later converted into a Python list for storage and comparison.

□ Image Encoder Output → Latent visual embedding □ Audio Encoder Output → Latent audio embedding
Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 54 □ Text Encoder Output → Latent textual embedding Example Text Encoder Output tensor([0042, -0118, 0305, , -0091]) After projection latent_key = [013, -007, 022, , 004] # 15D These representations are passed to the latent alignment and memory modules.

543 Memory Data Structure The system uses a persistent JSON-based memory store to retain multimodal experiences.

Each memory item is represented as a dictionary and stored in a list within a JSON file.

Each memory entry contains □ type Modality of the stored experience □ key 15-dimensional latent vector for similarity indexing □ value Semantic embedding for contextual understanding □ caption Human-readable description □ processor metadata Emotion, importance score, and processing details □ timestamp Implicit ordering based on insertion Example Memory Entry { type text, key [018, -004, 027, , 009], value [0311, -0154, 0208, , -0061], caption Einsteins mass-energy equivalence principle, processor { emotion neutral, importance 08 } } During retrieval, new query embeddings are compared against these stored keys and values.

544 Agent Reasoning Data Structure Agent outputs are represented as Python dictionaries, allowing different agents to return different types of outputs while maintaining a common structure.

Example Normal Agent Output { agent_type normal, reasoned_sentence The equation explains how mass can be converted into energy, facts [{ fact Energy is proportional to mass } Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 55], knowledge [{ concept Relativity, description Proposed by Einstein }], explanation { text Energy equals mass multiplied by the speed of light squared, audio

/static/audio/normal_audio_173821mp3 }, logs [Image processed, Audio processed, Memory retrieved] } This structure ensures traceability and transparency of reasoning.

545 Cross-Agent Feedback Data Structure For neuro-adaptive feedback, agent-to-agent translations are also stored as structured dictionaries.

Example Mute → Blind Feedback { from mute, to blind, audio_text Let me explain the formula step by step, audio_file /static/audio/blind_feedback_173899mp3 } This enables collaborative reasoning across agents.

546 Visualization Data Structures For visualization, semantic embeddings are temporarily stored as NumPy arrays and reduced using PCA.

Example embeddings = np.vstack([image_embedding, audio_embedding, text_embedding]) reduced = PCA(n_components=3).fit_transform(embeddings) The reduced coordinates are plotted and saved as image files for frontend rendering.

55 Algorithm Design This section presents the core algorithms that govern the internal operation of the Neuro-Adaptive Multi-Agent System.

The algorithms are expressed in clear, step-by-step pseudocode to explain how multimodal inputs are processed, how adaptive agents are selected, how memory is retrieved, and how final reasoning outputs are generated.

These algorithms collectively demonstrate the neuro-adaptive reasoning flow implemented in the system, from sensory input handling to explainable output generation.

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of CS&E, BIT 2025-26 56 551 Algorithm 1 Multimodal Input Processing To accept user inputs (image, audio, text), preprocess them, and prepare them for feature extraction.

Algorithm Input_Processing Input Image file (optional), Audio file (optional), Text input (optional) Output

Processed image, processed audio, cleaned text, modality flags Steps Receive input files and text from user interface.

If image is provided ☐ Load image.

☐ Resize and normalize image.

If audio is provided ☐ Convert audio to mono format.

☐ Resample audio to standard frequency.

☐ Apply voice activity detection.

If text is provided ☐ Clean and normalize text.

Identify which modalities are available.

Return processed inputs and modality availability information.

552 Algorithm 2 Feature Encoding To convert each available modality into numerical feature representations.

Algorithm Feature_Encoding Input Processed image, processed audio, cleaned text Output Image embedding, Audio embedding, Text embedding Steps If image exists ☐ Pass image through Image Encoder.

If audio exists ☐ Pass audio through Audio Encoder.

If text exists: Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 57 ☐ Pass text through Text Encoder.

Normalize all generated embeddings.

Return available embeddings.

553 Algorithm 3 Query Construction To construct structured query objects for each modality.

Algorithm Build_Query Input Embeddings from encoders, processed modality metadata Output List of modality query objects Steps Initialize empty query list.

For each available modality ☐ Create a dictionary containing o modality type o query text

(caption/transcript/text) o latent key embedding o semantic value embedding o metadata (emotion, importance)

Append modality entry to query list.

Return query list.

554 Algorithm 4 Memory Retrieval Purpose To retrieve relevant past experiences to support reasoning.

Algorithm Retrieve_Memory Input Query list, Memory store Output Top-N relevant memory entries Steps Load stored memory entries.

For each query modality ☐ Compute similarity between query and memory entries using Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 58 o semantic similarity o latent embedding similarity o keyword overlap o modality relevance Rank memory entries based on combined score.

Select top-N most relevant memories.

Return ranked memory list.

555 Algorithm 5 Agent Selection and Reasoning Purpose To perform adaptive reasoning based on available modalities.

Algorithm Agent_Reasoning Input Query list, Retrieved memory, Agent type Output Agent-specific reasoning output Steps Determine agent type ☐ Normal Agent → image + audio + text ☐ Blind Agent → audio + text ☐ Mute Agent → image + text Filter query entries based on agent modality constraints.

Combine relevant query information with retrieved memory.

Generate a reasoning sentence using reasoning engine.

Return agent reasoning output.

556 Algorithm 6 Explanation Generation Purpose To generate human-like explanations adapted to agent type.

Algorithm Generate_Explanation Input Agent reasoning output, facts, knowledge Output Final explanation (text and/or audio) Steps Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 59 Format reasoning context.

If agent is Normal ☐ Generate text explanation.

☐ Optionally generate audio explanation.

If agent is Blind ☐ Generate detailed audio explanation.

If agent is Mute ☐ Generate concise textual explanation.

Return formatted explanation.

557 Algorithm 7 Cross-Agent Feedback Purpose To translate reasoning between agents for neuro-adaptive

feedback.

Algorithm Agent_Feedback Input Source agent output, Target agent type Output Translated feedback output

Steps If source is Blind and target is Mute □ Convert spoken explanation into structured text.

If source is Mute and target is Blind □ Convert text explanation into spoken audio.

Return translated feedback.

558 Algorithm 8 Output Delivery Purpose To deliver final results to the user interface.

Algorithm Deliver_Output Input Final explanation, visualization data, logs Output User-visible response Steps

Package explanation, agent type, facts, and logs.

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of CS&E, BIT 2025-26 60 Generate visualization references if requested.

Send response to frontend.

Store reasoning instance in memory for future use.

End interaction.

CHAPTER 6 IMPLEMENTATION Dept.

of CS&E, BIT 2025-26 61 CHAPTER 6 IMPLEMENTATION 61 Implementation Approach The Neuro-

Adaptive Multi-Agent System is implemented using a modular, layered software architecture that enables adaptive multimodal reasoning under varying sensory conditions.

The implementation focuses on practicality, clarity, and robustness, ensuring that the system can operate effectively even when one or more input modalities are unavailable.

The overall approach follows a sequential processing pipeline that begins with multimodal input acquisition, proceeds through encoding and agent-based reasoning, and ends with explanation generation and visualization through a web interface.

The system is developed primarily using Python, with Flask serving as the application framework to integrate backend intelligence with a user-friendly frontend interface.

611 Multimodal Input Acquisition The system supports three types of user inputs image, audio, and text, reflecting real- world scenarios where information is received through multiple sensory channels.

Image inputs are uploaded through the web interface and stored temporarily on the server.

Basic preprocessing operations such as resizing and format standardization are performed to ensure compatibility with the image encoding pipeline.

Audio inputs are accepted in common formats such as WAV and MP3.

Uploaded audio files are converted into a standardized format using an audio processing library.

The conversion ensures mono-channel audio and a uniform sampling rate, which improves the reliability of downstream speech processing and feature extraction.

Text input is directly entered by the user through the interface.

Since the text encoder internally manages tokenization and normalization, only minimal preprocessing is applied at this stage.

This flexible input handling mechanism allows the system to function correctly even when users provide only one or two modalities, which is essential for adaptive and assistive applications.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 62 612 Multimodal Feature Encoding After input acquisition, each available modality is processed using a dedicated encoder that converts raw data into numerical feature representations.

The image encoder extracts visual features that represent objects, structures, and visual patterns present in the image.

The audio encoder captures meaningful characteristics from speech or sound signals, focusing on temporal and spectral information.

The text encoder converts user-provided text into dense semantic embeddings that represent contextual meaning.

All encoders produce embeddings that are compatible with further processing and comparison.

This encoding stage transforms heterogeneous inputs into structured numerical forms, enabling consistent reasoning across different modalities.

613 Neuro-Adaptive Agent Layer The core adaptive behavior of the system is achieved through a multi-agent design.

Three agents are implemented, each simulating a different sensory condition □ The Normal Agent performs

reasoning using all available modalities.

- The Blind Agent operates without visual input and relies on audio and text information.

- The Mute Agent ignores audio input and reasons using image and text information.

The system automatically selects the appropriate agent based on the modalities provided by the user.

This mechanism allows the system to continue functioning even when certain sensory inputs are missing, thereby reflecting neuro-adaptive behavior inspired by human cognition.

614 Reasoning and Cognitive Processing Once the agent is selected, the system performs reasoning by combining current input representations with relevant contextual information.

The reasoning process focuses on understanding relationships, extracting meaning, and forming coherent explanations rather than merely describing raw input content.

The reasoning pipeline integrates □ semantic interpretation of inputs, □ contextual linking between modalities, □ structured reasoning generation.

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of CS&E, BIT 2025-26 63 This stage ensures that the output reflects interpreted meaning rather than simple pattern recognition.

615 Memory Storage and Retrieval To support context-aware reasoning, the system includes a memory module that stores previous interactions and reasoning outcomes.

Each interaction contributes a compact representation to memory, which can be retrieved later when similar queries are encountered.

Memory retrieval is based on similarity matching between current inputs and stored experiences.

This allows the system to enhance explanations using prior knowledge and maintain continuity across interactions, without requiring retraining of models.

616 Output Generation and Visualization The final stage of the pipeline generates outputs in a form suitable for the active agent.

Depending on the agent type, the output may be provided as □ textual explanations, □ audio explanations, □ or a combination of both.

In addition to explanations, the system supports visualization of embeddings and reasoning spaces using dimensionality reduction techniques.

These visualizations help illustrate how different modalities and agents behave under varying sensory conditions.

617 Flask-Based System Integration Flask serves as the central integration layer connecting user interactions with backend processing.

It manages □ handling of HTTP requests, □ file uploads, □ execution of agent reasoning, □ delivery of results in structured JSON format, □ rendering of HTML templates, □ visualization routing.

This integration allows the system to function as an interactive web-based application, making it accessible, responsive, and suitable for demonstration and evaluation.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 64 62 Coding Details And Code Efficiency 621 Code Organization and Modularity The Neuro-Adaptive Multi-Agent System is implemented using a modular Python package structure, where each directory represents a logically independent subsystem.

This modular organization improves code readability, simplifies debugging, enables independent development of components, and supports future extensibility.

Each module performs a well-defined role in the multimodal reasoning pipeline while interacting with other modules through clearly defined interfaces.

The major components of the codebase are described below along with representative code snippets.

processors/ This module handles modality-specific preprocessing, converting raw user inputs into structured information suitable for neural encoding and reasoning.

```
image_processor.py – Image emotion and caption extraction from deepface
import DeepFace from
models
image_captioning import generate_caption
def process_image(image_path)
analysis = DeepFace.analyze(
img_path=image_path, actions=[emotion], enforce_detection=False )
emotion = analysis[0][dominant_emotion]
caption = generate_caption(image_path)
return { emotion:emotion, caption:caption }
audio_processor.py – Audio
conversion, speech detection, transcription from pydub
import AudioSegment
Neuro Adaptive AI Agents Dept.
of CS&E, BIT 2025-26 65
from models
speech_to_text import transcribe_audio
def process_audio(audio_path)
```

audio = AudioSegmentfrom_file(audio_path) audio = audioset_channels(1)set_frame_rate(16000)
 audioexport(convertedwav, format=wav) transcript = transcribe_audio(convertedwav) return {transcribed_text
 transcript} text_processorpy – Emotion and text analysis from modelsemotion_recognition import
 detect_text_emotion def process_text(text) emotion, score, importance = detect_text_emotion(text) return {
 emotion emotion, importance importance, summary text } models/ This directory contains trained neural models
 and supporting utilities used across the system, including multimodal encoders and emotion/speech models.
 image_embedding = image_encoder(image_tensor) audio_embedding = audio_encoder(audio_tensor)
 text_embedding = text_encoder(text_tensor) Speech-to-text and emotion recognition models are also accessed
 internally by the processors.
 memory/ This module implements persistent memory storage and similarity-based retrieval to support memory-
 augmented reasoning.
 memory_managerpy – Persistent JSON-based memory storage import jsonNeuro Adaptive AI Agents Dept.
 of CS&E, BIT 2025-26 66 def add_to_memory(entry, file_path) memory = jsonload(open(file_path))
 memoryappend(entry) jsondump(memory, open(file_path, w), indent=2) retrievalpy – Similarity-based memory
 ranking from utilssimilarity import cosine_sim def retrieve_topn(query_key, memory, top_n=5) scores = [(m,
 cosine_sim(query_key, m[key])) for m in memory] scoressort(key=lambda x x[1], reverse=True) return
 scores[:top_n] agents/ This module contains logic for Normal, Blind, and Mute agents, each designed to operate
 under different sensory constraints.
 if agent_type == normal active_modalities = [image, audio, text] elif agent_type == blind active_modalities =
 [audio, text] elif agent_type == mute active_modalities = [image, text] Each agent ignores unavailable modalities
 and performs reasoning based only on permitted sensory inputs.
 reasoning/ This module handles query understanding, cognitive analysis, memory-enhanced reasoning, and
 explanation generation.
 query_info = classify_query(query_text) analysis = enhanced_cognitive_processing(query_entry,
 retrieved_memory) explanation = generate_explanation(query_text, analysis) core/ The core module coordinates
 multimodal query construction and manages the overall Neuro Adaptive AI Agents Dept.
 of CS&E, BIT 2025-26 67 reasoning flow.
 query = build_query(image_path=image_path, audio_path=audio_path, text_input=text_input) Flask
 Application Files The Flask framework is used as the integration and interaction layer between the user interface
 and the backend neuro-adaptive reasoning system.
 It acts as a lightweight web server that coordinates multimodal input handling, agent invocation, visualization
 generation, and response delivery.
 Flask enables real-time interaction with the multi-agent architecture through a browser-based interface, making
 the system accessible without requiring command-line execution.
 The Flask layer performs the following key responsibilities □ Handling HTTP requests from the frontend □
 Managing file uploads for image and audio modalities □ Validating user inputs and agent selection □ Triggering
 backend reasoning workflows □ Returning structured JSON responses □ Serving generated audio files and
 visualization images Route for System Entry and Interface Rendering The initial routes are responsible for
 loading the user interface and guiding the user into the system workflow.
 @app.route(/) def home() return render_template(landinghtml) @app.route(/start) def start_system() return
 render_template(index2html) These routes render the landing page and the main multi-agent interaction
 interface.
 Neuro Adaptive AI Agents Dept.
 of CS&E, BIT 2025-26 68 Route for Multimodal File Upload and Input Handling Flask handles uploaded image
 and audio files using secure file handling mechanisms to prevent invalid formats and ensure safe storage.
 if image_file and allowed_file(image_filefilename) image_path= ospathjoin(UPLOAD_FOLDER,
 secure_filename(image_filefilename)) image_filesave(image_path) if audio_file and
 allowed_file(audio_filefilename) audio_path = ospathjoin(UPLOAD_FOLDER,
 secure_filename(audio_filefilename)) audio_filesave(audio_path) This ensures that uploaded media is validated,
 securely stored, and correctly passed to the backend processing pipeline.
 Route for 3D Multimodal Input Visualization Flask also supports visualization requests, where embeddings are
 projected into a 3D space for interpretability.

@app.route(/generate_3d, methods=[POST]) def generate_3d() # Encode image, audio, and text embeddings = np.vstack([img_vec, aud_vec, txt_vec]) reduced = PCA(n_components=3).fit_transform(embeddings) plot_url = save_plot(reduced) return jsonify({redirect_url: plot_url}) This route generates PCA-based visualizations only when explicitly requested, ensuring visualization remains separate from reasoning execution.

Route for Reasoning Space Visualization The system supports reasoning comparison across agents using embedding visualization.

@app.route(/visualize_reasoning, methods=[POST]) def visualize_reasoning() embeddings = model.encode([normal_r, blind_r, mute_r]) reduced = PCA(n_components=3).fit_transform(embeddings) plot_url = save_plot(reduced) return jsonify({redirect_url: plot_url}) This enables comparative analysis of how different agents reason under sensory

Neuro Adaptive AI Agents Dept.
of CS&E, BIT 2025-26 69 constraints.

Response Handling and Output Delivery All backend results are returned as structured JSON responses, allowing the frontend to dynamically display explanations, audio outputs, and visual content.

return jsonify({ agent_type: agent_type, reasoned_sentence: reasoning, explanation: explanation, logs: execution_logs }) Figure 61 Sequence Diagram of User Interaction and Flask-Based Processing Flow 622

Device Handling and Model Execution The Neuro-Adaptive Multi-Agent System is designed to execute on both CPU and GPU environments.

During system initialization, the device is automatically selected based on hardware availability.

If a CUDA-enabled GPU is present, the system uses GPU acceleration; otherwise, it safely falls back to CPU execution.

This design ensures portability across different machines, including academic laboratory systems where GPU availability may vary.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 70 import torch DEVICE = torch.device('cuda' if torch.cuda.is_available() else 'cpu')

img_enc = ImageEncoder().to(DEVICE) aud_enc = AudioEncoder().to(DEVICE) txt_enc =

TextEncoder().to(DEVICE) This approach guarantees stable execution without enforcing hardware dependency.

623 Modality-Specific Processing Efficiency The system processes only those modalities that are explicitly provided by the user.

If a modality is missing, its processing pipeline is skipped entirely.

This conditional execution avoids unnecessary computation and improves runtime efficiency.

if image_path: image_query = process_image(image_path) if audio_path: audio_query = process_audio(audio_path) if text_input: text_query = process_text(text_input) By selectively activating pipelines, the system maintains efficiency while preserving correct multimodal behavior.

624 Audio Processing Reliability Audio inputs are normalized to ensure consistent analysis across different recording environments.

Uploaded audio files are converted to a standard format before further processing.

Voice Activity Detection (VAD) is applied to determine whether meaningful speech is present.

Speech-to-text transcription is performed only when speech is detected, preventing unnecessary processing of noise or silence.

audio = AudioSegment.from_file(audio_path) audio = audio.set_channels(1).set_frame_rate(16000) Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 71 audio.export(convertedwav, format='wav') if is_speech(convertedwav): transcript = transcribe_audio(convertedwav) else: transcript = This pipeline increases robustness and improves reasoning accuracy.

625 Memory Storage and Retrieval Implementation The system maintains a persistent memory using a JSON-based storage mechanism.

Each interaction is stored as a structured dictionary containing embeddings, metadata, and timestamps.

Memory retrieval is based on similarity measures and relevance scoring.

memory_entry = { type: 'text', key: latent_key, value: semantic_embedding, caption: summary, timestamp: datetime.now() } memory_manager.add_to_memory(memory_entry) Example results = retriever.retrieve_topn(query_list, top_n=5) This approach is efficient and sufficient for an academic-scale system.

626 Agent-Based Conditional Reasoning The system activates a single agent at a time based on user selection

and modality availability.

Each agent ignores unsupported modalities internally, ensuring correct neuro-adaptive behavior.

```
if agent_type == normal active_modalities = [image, audio, text] elif agent_type == blind active_modalities = [audio, text] elif agent_type == mute
```

Neuro Adaptive AI Agents Dept.
of CS&E, BIT 2025-26 72 active_modalities = [image, text] This design prevents redundant computation and preserves agent-specific reasoning constraints.

627 Visualization-Specific Computation Dimensionality reduction using PCA is performed exclusively for visualization purposes.

The reduced embeddings are never used in reasoning or memory retrieval, ensuring that visualization does not affect cognitive accuracy.

```
from sklearn.decomposition import PCA pca = PCA(n_components=3) reduced_embeddings = pca.fit_transform(embeddings)
```

Visualization is generated only when explicitly requested by the user.
628 Flask Runtime Behavior Flask serves as the middleware layer connecting users to backend cognitive components.

It handles input validation, agent execution control, and response formatting.

```
@app.route('/run_agent', methods=[POST]) def run_agent() result = run_multi_agent_query(agent_type=request.form.get('agent_type'), image_path=image_path, audio_path=audio_path, text_input=text_input ) return jsonify(result)
```

This ensures responsive interaction without redundant backend execution.
629 Error Handling and Fail-Safe Execution To guarantee uninterrupted execution, all major processing blocks are wrapped in exception handling.

When an error occurs, the system returns a safe response instead of terminating execution.

Neuro Adaptive AI Agents Dept.

```
of CS&E, BIT 2025-26 73 try result = process_audio(audio_path) except Exception as e result = {error: Audio processing failed}
```

This fail-safe design ensures reliability even when inputs are incomplete or unexpected.
CHAPTER 7 TESTING Dept.

of CS&E, BIT 2025-26 74 CHAPTER 7 TESTING 71 Testing Approach Testing was carried out systematically to verify the correctness, reliability, and stability of the proposed Neuro-Adaptive Multi-Agent System.

The testing process followed an incremental strategy, starting with unit testing of individual modules and progressing toward integrated testing of the complete system.

Special attention was given to validating modality-specific processing, memory operations, agent behavior under sensory constraints, and visualization outputs.

711 Unit Testing Unit testing was performed to validate each module independently before system integration.

Each functional component was tested using controlled inputs to ensure correctness in isolation.

7111 Modality Input Validation Objective To verify correct processing of individual input modalities (image, audio, and text).

Input Modality	Test Description	Expected Output	Observed Output	Accuracy (%)
Image	Person riding bicycle	Caption + emotion	Correct caption, emotion = happy	95%
Audio	“Einstein discovered mass-energy equivalence”	Transcription + emotion	Accurate transcription, neutral emotion	93%
Text	“This was unexpected”	Emotion + importance	Surprise emotion detected	98%

Table 71 Modality Input Testing
Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 75 7112 Encoder Output Projection Testing Objective To verify dimensionality reduction from encoder outputs to the shared latent space.

Test Case	Encoder Type	Input Dimension	Expected Output	Observed Output	Accuracy (%)
Image	Encoder 256D	15D vector (15,) normalized	100%	100%	100%
Audio	Encoder 256D	15D vector (15,) normalized	100%	100%	100%
Text	Encoder 256D	15D vector (15,) normalized	100%	100%	100%

Table 72 Encoder Projection Testing
7113 Query Builder Validation Objective To ensure correct formation of query objects used for memory and reasoning.

Test Case	Query Builder Modality	Expected Fields	Observed Structure	Accuracy (%)
Image	type, key, value, metadata	All fields present	100%	100%
Text	type, key, value, metadata	Correct mapping	100%	100%

Table 73 Query Builder Testing
7114 Memory Storage Testing Objective To verify correct storage of new memory entries without overwriting previous data.

Test Case	Memory Storage Operation	Expected Result	Observed Result	Accuracy (%)
Add entry	New			

entry appended JSON size increased 100% Timestamp Auto-generated Correct timestamp 100% Table 74 Memory Storage Testing 7115 Memory Retrieval Testing Objective To validate Top-N retrieval based on similarity scores.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 76 Test Case Table Memory Retrieval Query Type Expected Result Observed Result Accuracy (%) Text query Relevant Top-N memories Correct ranking 92% Audio query Semantic match Valid scores 90% Table 75 Memory Retrieval Testing 7116 Agent-Level Reasoning Testing Objective To verify correct reasoning behavior for each agent.

Test Case Table Agent Testing Agent Type Enabled Modalities Expected Behavior Accuracy (%) Normal Image + Audio + Text Full reasoning 96% Blind Audio + Text No image dependency 94% Mute Image + Text No audio dependency 93% Table 76 Agent Reasoning Testing 7117 PCA Visualization Testing Objective To ensure correct dimensionality reduction and modality separation.

Test Case Table PCA Visualization Visualization Type Expected Outcome Accuracy (%) Input PCA Separate axes for modalities 95% Reasoning PCA Distinct agent positions 94% Table 77 PCA Visualization Testing 7118 PCA-Based Modality Visualization Objective To verify that multimodal embeddings (image, audio, and text) are properly separated in a three-dimensional latent space after dimensionality reduction using Principal Component Analysis (PCA).

Testing Activities ☐ Apply PCA to high-dimensional modality embeddings ☐ Reduce embeddings to three principal components Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 77 ☐ Generate and inspect 3D scatter plots ☐ Observe relative spatial positioning of modalities and agent behavior Expected Behaviour ☐ Image embeddings should predominantly align along the X-axis ☐ Audio embeddings should predominantly align along the Y-axis ☐ Text embeddings should predominantly align along the Z-axis ☐ Blind agent reasoning should lie on the Y-Z plane ☐ Mute agent reasoning should lie on the X-Z plane Observed Results ☐ PCA output dimension (3, 3) ☐ Explained variance ratio 091 ☐ Clear spatial separation between modalities observed Visualization Interpretation ☐ Blue points represent image embeddings ☐ Green points represent audio embeddings ☐ Red points represent text embeddings ☐ Distinct clustering confirms effective modality separation ☐ Agent-specific reasoning follows expected neuro-adaptive planes Accuracy Assessment Evaluation Aspect Observed Behaviour Accuracy (%) Modality separation Clear spatial distinction 95% Axis alignment Correct axis dominance 94% Agent plane behaviour Blind & Mute follow constraints 93% Table 78 Accuracy Assessment of PCA-Based Modality Visualization Conclusion The PCA-based visualization confirms that the system successfully aligns multimodal embeddings into a structured latent space.

The observed separation and agent-specific positioning validate the effectiveness of the multimodal embedding design and neuro- adaptive reasoning behavior.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 78 712 Integrated Testing Integrated testing was performed after successful completion of unit testing to ensure that all subsystems multimodal encoders, query builders, memory components, reasoning agents, and the reasoning engine operate correctly when combined.

This phase validates end-to-end execution, data consistency across modules, and coherent agent-based reasoning under different sensory conditions.

7121 Multimodal Fusion Pipeline Testing Objective To verify end-to-end fusion of image, audio, and text streams into a unified reasoning pipeline.

Testing Activities ☐ Provide image, audio, and text inputs simultaneously ☐ Validate preprocessing → encoding → projection → query formation ☐ Test behavior when one or more modalities are missing Observed Outputs ☐ Unified query object containing 15D latent keys, 384D semantic values, and modality metadata ☐ Dynamic switching between Normal, Blind, and Mute configurations Evaluation Aspect Observed Behaviour Accuracy (%) Full-modality fusion Correct tri-modal alignment 94% Missing-modality handling Correct agent switching 93% Query consistency No data mismatch 95% Table 79 Multimodal Fusion Pipeline Accuracy 7122 Memory Retrieval and Reasoning Integration Testing Objective To ensure that memory retrieval results correctly influence reasoning across all agents.

Testing Activities ☐ Execute reasoning with empty memory (fallback reasoning) ☐ Insert sample memory entries and test retrieval ranking ☐ Verify memory content integration into reasoning output Observed Outputs ☐

Ranked memory items with similarity scores □ Reasoning enriched with retrieved factual and contextual knowledge
Neuro Adaptive AI Agents Dept.
of CS&E, BIT 2025-26 79 Evaluation Aspect Observed Behaviour Accuracy (%) Retrieval relevance Correct Top-N ranking 92% Knowledge grounding Memory facts used in reasoning 90% Agent consistency Correct modality-restricted reasoning 93% Table 710 Memory–Reasoning Integration Accuracy 7123 Cognitive Layer and Memory Compression Integration Objective To validate interaction between cognitive analysis and memory summarization mechanisms.

Testing Activities □ Generate multiple high-volume reasoning queries □ Validate summary generation and confidence scoring □ Observe compression behavior over time Observed Outputs □ Summarized memory entries with confidence and timestamps □ Reduction in memory size after compression Evaluation Aspect Observed Behaviour Accuracy (%) Summary stability Consistent semantic summaries 91% Confidence scoring Reliable confidence values 90% Compression effectiveness Reduced redundancy 89% Table 711 Cognitive Compression Accuracy 7124 Multi-Agent Interaction Testing (Normal ↔ Blind ↔ Mute) Objective To verify cross-agent feedback and modality translation between agents.

Testing Activities □ Blind agent reasoning → verify text feedback for Mute agent □ Mute agent reasoning → verify audio feedback for Blind agent □ Validate semantic consistency across conversions Observed Outputs □ Spoken reasoning converted to structured text □ Text reasoning converted to natural audio explanations
Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 80 Interaction Type Observed Behaviour Accuracy (%) Blind → Mute Accurate text extraction 92% Mute → Blind Clear audio synthesis 91% Semantic preservation Meaning retained 90% Table 712 Cross-Agent Interaction Accuracy 7125 PCA Visualization Integration Testing Objective To verify correctness of PCA-based visualization for inputs and reasoning outputs.

Testing Activities □ Generate PCA plots before reasoning (modalities) □ Generate PCA plots after reasoning (agents) □ Observe spatial separation and clustering Observed Outputs □ Modality clusters aligned with expected axes □ Agent reasoning vectors positioned according to modality constraints Evaluation Aspect Observed Behaviour Accuracy (%) Modality clustering Clear separation 95% Agent positioning Correct reasoning planes 93% Visualization clarity Interpretable plots 94% Table 713 PCA Visualization Accuracy 7126 Full System Execution Testing (End-to-End) Objective To validate the complete workflow from user input to final reasoning and visualization.

Testing Activities □ Upload multimodal inputs through Flask UI □ Trigger visualization and agent reasoning □ Inspect frontend outputs, logs, and memory updates Observed Outputs □ Structured JSON responses returned to frontend □ Audio and text explanations rendered correctly □ Memory updated after each reasoning cycle
Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 81 Evaluation Aspect Observed Behaviour Accuracy (%) Input-to-output flow Successful execution 94% Frontend response Correct rendering 93% Memory update Persistent storage 92% Table 714 End-to-End System Accuracy Conclusion of Integrated Testing Integrated testing confirms that all system components function cohesively under real- world execution conditions.

The system demonstrates reliable multimodal fusion, adaptive agent reasoning, memory-guided inference, and interpretable visualization with consistently high accuracy across testing scenarios.

CHAPTER 8 RESULTS DISCUSSION AND PERFORMANCE ANALYSIS Dept.

of CS&E, BIT 2025-26 82 CHAPTER 8 RESULTS DISCUSSION AND PERFORMANCE ANALYSIS 81

Results This section presents the experimental results obtained from the Neuro-Adaptive Multimodal AI System. The results focus on how multimodal inputs (image, audio, and text) are represented in a shared semantic space, how their combined meaning evolves over time, and how semantic drift can be visually and numerically analyzed using dimensionality reduction techniques.

Principal Component Analysis (PCA) is used to project high-dimensional embeddings into interpretable 3D and 2D spaces.

811 3D Projection of Multimodal Inputs To analyze how different input modalities are internally represented, sentence-level embeddings from image captions, audio transcriptions, and text queries were projected into a common 3D semantic space using PCA.

Objective The objective of this experiment is to verify that □ Each modality occupies a distinct region in the semantic space □ Related inputs across modalities maintain semantic proximity □ Multimodal representations

are separable yet aligned Experimental Setup □ Image captions, audio transcripts, and text queries were encoded using a transformer-based embedding model.

□ All embeddings were reduced from 384 dimensions to 3 dimensions using PCA.

□ Each modality was plotted independently without inter-point connections to avoid artificial correlation.

Observed Results □ Blue points (Image modality) clustered along a consistent region, reflecting visual-semantic coherence.

□ Green points (Audio modality) formed a separate but nearby cluster, indicating alignment with spoken explanations.

□ Red points (Text modality) showed stronger variation, reflecting higher linguistic abstraction and query intent diversity.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 83 □ Overlapping regions between modalities indicate successful multimodal semantic alignment.

Interpretation The 3D visualization confirms that □ The system learns modality-aware embeddings □ Different sensory inputs remain distinguishable while sharing a common semantic frame □ This separation is critical for agent specialization (Normal, Blind, Mute agents) This validates the encoder and latent projection design described in earlier chapters.

8111 2D Semantic Drift Analysis of Multimodal Queries To analyze how multimodal understanding evolves across multiple inputs, a semantic drift experiment was conducted using centroid representations.

Objective To study □ Temporal or contextual evolution of meaning □ Stability and directional change in semantic interpretation □ Aggregated multimodal behavior over multiple inputs Methodology □ For each input

instance, image, audio, and text embeddings were averaged to form a semantic centroid.

□ Centroids were projected into a 2D space using PCA.

□ A dashed line was used to visualize semantic drift across inputs.

Input ID X-Coordinate Y-Coordinate Input-1 -02624 00688 Input-2 -01896 -02191 Input-3 01344 04628 Input-4 -01888 -01043 Input-5 05064 -02081 Table 81 Observed Centroid Coordinates of Input Embeddings

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 84 Figure 81 3D Projection of Multimodal Inputs Figure 82 2D Semantic Drift of Multimodal Inputs

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 85 PCA Variance Explanation PC1 4037% PC2 3125% Total Variance Captured 7162%

Key Observations □ Early inputs cluster closer together, indicating similar conceptual grounding.

□ Later inputs move significantly along PC1, reflecting topic expansion (from basic explanation to astrophysics).

□ The trajectory clearly shows semantic drift, not random noise.

□ PC1 captures conceptual progression, while PC2 reflects abstraction depth.

Interpretation The semantic drift curve demonstrates that □ The system does not treat inputs independently □

Meaning evolves consistently as new contextual information is introduced □ Multimodal fusion produces stable yet adaptable representations This behavior closely resembles human conceptual evolution, validating the neuro-adaptive design philosophy.

8112 Discussion on Semantic Consistency and Adaptability The combined 3D and 2D results indicate that □

Multimodal embeddings remain consistent across modalities □ Semantic centroids provide a reliable summary of system understanding □ Drift is meaningful and interpretable, not arbitrary This confirms that the system □

Successfully integrates multimodal inputs □ Preserves long-term semantic coherence □ Supports adaptive reasoning across time and context 812 Top-N Memory Retrieval Results and Accuracy Analysis This subsection analyzes the performance of the Top-N memory retrieval mechanism of the proposed Neuro-Adaptive Multi-Agent System.

The evaluation focuses on how effectively the system retrieves semantically relevant knowledge from long-term memory when presented with linguistically varied but conceptually related queries.

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of CS&E, BIT 2025-26 86 Objective The primary objectives of this evaluation are □ To measure retrieval accuracy across multiple natural language queries □ To analyze semantic alignment between user queries and retrieved memory entries □ To examine the stability of retrieval scores under different query formulations

Experimental Setup □ Three user queries related to the Law of Conservation of Energy were provided as input.
□ Each query differed in structure, length, and linguistic formulation while maintaining the same conceptual intent.

□ The memory retrieval module computed o Latent key similarity (sim_key) o Semantic similarity (sim_value)
o Final retrieval score (weighted combination) □ Mean accuracy and standard deviation were calculated from the similarity scores of the Top-N retrieved memories.

□ Retrieval accuracy was visualized using a bar graph for comparative analysis.

Query-Wise Retrieval Results Query 1 Input Query “Hello, can you explain the law of conservation of energy? Please describe it in simple words” Top-1 Retrieved Memory □ Caption Explanation of the law of conservation of energy and its transformations □ Retrieval Score 08838 □ Semantic Similarity (sim_value) 07675 □ Latent Key Similarity (sim_key) 09999 Accuracy Metrics □ Mean Accuracy 07803 □ Standard Deviation 00

Discussion This query achieved a moderate-to-high accuracy, indicating strong semantic alignment despite informal phrasing.

The high latent similarity confirms that the system correctly Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 87 mapped the conceptual intent to the stored memory.

Query 2 Input Query “First define the law of conservation of energy.

Next, explain why energy cannot be created or destroyed” Top-1 Retrieved Memory □ Caption Formal definition with explanatory context □ Retrieval Score 08175 □ Semantic Similarity (sim_value) 07341 □ Latent Key Similarity (sim_key) 09010 Accuracy Metrics □ Mean Accuracy 07228 □ Standard Deviation 00

Discussion This query achieved the highest overall accuracy among all test cases.

The structured and sequential nature of the query enabled precise semantic matching and efficient memory retrieval.

Query 3 Input Query “What does the law of conservation of energy mean? Summarize the concept in one sentence” Top-2 Retrieved Memories Conceptual summary explanation o Score 07753 o sim_value 06844 o sim_key 08662 Mathematical formulation-based explanation o Score 07150 o sim_value 06535 o sim_key 07765 Accuracy Metrics □ Mean Accuracy 06265 □ Standard Deviation 00170 Neuro Adaptive AI Agents Dept. of CS&E, BIT 2025-26 88 Discussion This query produced multiple relevant memory entries but showed slightly reduced accuracy due to the high-level summarization request.

The presence of variance reflects broader semantic dispersion during retrieval.

Query Mean Accuracy Performance Interpretation Query 1 ~063 Moderate semantic alignment Query 2 ~078

Highest accuracy and coherence Query 3 ~072 Stable retrieval with minor variance Table 82 Accuracy

Comparison Summary Graphical Interpretation The accuracy comparison plot demonstrates that □ Structured queries yield higher retrieval accuracy □ Informal or compressed queries slightly reduce precision □ Overall retrieval remains robust across linguistic variations Key Observations □ The hybrid similarity strategy effectively balances semantic and latent matching □ Keyword boosting and modality weighting improve ranking accuracy □ Retrieval performance remains stable under query reformulation □ The system consistently retrieves conceptually correct knowledge 813 Reasoning Behaviour Across Multimodal Inputs This section evaluates the reasoning behaviour of the three neuro-adaptive agents Normal, Blind, and Mute when exposed to heterogeneous multimodal inputs.

The objective of this analysis is to understand how each agent adapts its reasoning strategy under varying conditions of modality availability and semantic alignment, reflecting the system’s neuroplastic design.

Four carefully designed test cases were constructed to assess system robustness under the following scenarios □

Semantically mismatched modalities □ Missing modality conditions □ Partially aligned multimodal inputs

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 89 □ Fully aligned multimodal inputs Each agent generated an independent reasoning output based solely on its available sensory channels.

The resulting outputs were qualitatively analyzed to compare semantic consistency, failure modes, and compensatory behaviour.

8131 Case 1 Semantically Mismatched Inputs Input Configuration □ Image A dog playing in the grass □ Audio Definition of a neuro-adaptive system □ Text Query “Explain what it meant? really” Observed Behaviour In this scenario, all three modalities carried unrelated semantic content, resulting in strong cross-modal conflict.

□ The Blind Agent, which relies exclusively on audio and text, successfully generated a coherent explanation of

neuro-adaptive systems.

- The Normal Agent failed to produce valid reasoning due to conflicting signals entering the multimodal fusion pipeline.

- The Mute Agent also failed, as the irrelevant visual input dominated its reasoning process.

Interpretation This case demonstrates that under strong semantic mismatch, audio–text alignment dominates reasoning stability, while tri-modal fusion can degrade reasoning quality when modalities lack shared semantic grounding.

8132 Case 2 Missing Visual Modality Input Configuration □ Audio Request to explain Newton’s Third Law □ Text Query “Explain Newton third law?” □ Image Not provided Observed Behaviour □ Both the Normal Agent and Blind Agent produced accurate and semantically aligned explanations.

- The Blind Agent demonstrated stable reasoning by relying on linguistic and auditory cues.

- The Mute Agent correctly skipped reasoning due to the absence of its primary Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 90 modality (vision).

Interpretation This case validates the system’s neuro-adaptive compensation capability.

When a modality is missing, reasoning reorganizes around available inputs without semantic collapse, closely mirroring biological neuroplastic behaviour.

8133 Case 3 Partially Aligned Multimodal Inputs Input Configuration □ Image Caption Abstract energy diagram

- Audio Transcript Explanation of $E = mc^2$ □ Text Query “Explain why mass is multiplied with c ?” Observed Behaviour All three agents produced meaningful and convergent reasoning centered on mass– energy equivalence □ The Normal Agent generated the most comprehensive explanation by integrating visual abstraction with linguistic information.

- The Blind Agent emphasized theoretical interpretation based on audio–text semantics.

- The Mute Agent inferred reasoning from visual structure and textual cues.

Interpretation This confirms that the 15-dimensional latent key combined with 384-dimensional semantic value embeddings preserves sufficient shared structure to support stable reasoning even when modalities are only partially aligned.

8134 Case 4 Fully Aligned Multimodal Inputs Input Configuration □ Image Caption Abstract energy diagram □

Audio Transcript Definition of $E = mc^2$ □ Text Query “Explain this formula?” Observed Behaviour □ All three agents generated high-quality and consistent reasoning outputs.

- The Normal Agent effectively integrated visual cues with conceptual meaning.

- The Blind Agent focused on physical equivalence and theoretical explanation.

- The Mute Agent successfully inferred reasoning from image–text alignment despite the absence of audio cues.

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of CS&E, BIT 2025-26 91 Interpretation This scenario demonstrates maximum reasoning stability under ideal multimodal alignment and validates the effectiveness of the full multimodal fusion architecture.

8135 Summary of Reasoning Behaviour Across the four test cases, three consistent patterns emerged □ Semantic Coherence is Critical The Normal Agent performs strongly when modalities are aligned but is sensitive to cross-modal semantic conflict.

- Blind Agent is the Most Stable By relying on linguistically rich audio–text inputs, the Blind Agent consistently produces coherent reasoning even under partial mismatch.

- Mute Agent Behaves Conservatively It refuses reasoning when visual cues are absent and produces abstract explanations when audio is missing, demonstrating correct modality awareness.

Overall, these observations confirm that the neuro-adaptive agents exhibit modality- dependent behaviour, dynamic compensation, and robust reasoning under incomplete or noisy inputs.

The reasoning process closely mirrors the intended neuroplasticity- inspired design, reorganizing cognitive pathways based on modality availability.

8136 3D Reasoning Space Visualization To further analyze semantic differences in agent reasoning, the reasoning embeddings generated by the Normal, Blind, and Mute agents were projected into a shared three-dimensional PCA space.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 92 Figure 83 3D PCA visualization of reasoning embeddings across four multimodal

inputs.

8137 PCA-Based Reasoning Projection □ Reasoning embeddings were computed for all four evaluation inputs.

□ Principal Component Analysis (PCA) reduced high-dimensional embeddings into a 3D space.

□ Each input's agent embeddings were visualized together for direct comparison.

Figure 83 presents a four-panel collage showing PCA projections for Inputs 1 through 4. Each panel illustrates the relative positioning of the Normal, Blind, and Mute agent reasoning embeddings.

8138 Reasoning Proximity Analysis To quantify semantic similarity, Euclidean distances were computed between the Normal Agent and the two sensory-deprived agents.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 93 Input Normal–Blind Distance Normal–Mute Distance Closer Agent 1 13939 00337

Mute 2 07961 14343 Blind 3 05099 04867 Mute 4 08075 09140 Blind Table 83 Reasoning Proximity Analysis

Across All Inputs 8139 Interpretation The proximity analysis reveals that □ In some cases, the Normal Agent aligns more closely with the Mute Agent.

□ In other cases, it aligns more closely with the Blind Agent.

□ Semantic proximity depends on which modality is missing and the nature of the reasoning task, rather than consistently favoring one agent.

This confirms that reasoning similarity is context-dependent, further validating the adaptive and neuroplastic behaviour of the proposed multimodal reasoning system.

814 Blind Agent Semantic Drift Analysis This section analyzes the reasoning stability and semantic consistency of the Blind Agent, which operates without visual input and relies exclusively on audio–text modalities.

The objective is to examine how closely the Blind Agent's reasoning representations align with the original input semantics and to identify instances of semantic drift under varying input conditions.

To achieve this, both input embeddings and Blind Agent reasoning embeddings were projected into a shared three-dimensional PCA space, enabling direct geometric comparison between the original multimodal semantics (excluding vision) and the generated reasoning representations.

8141 3D Semantic Drift Visualization Figure 84 illustrates the 3D PCA alignment between the Blind Agent's input embeddings and its corresponding reasoning embeddings.

□ Circular markers (●) represent projected input embeddings.

□ Triangular markers (▲) represent projected Blind Agent reasoning embeddings.

□ Semi-transparent ellipsoids indicate local semantic neighborhoods, capturing the variance and dispersion of each embedding cluster.

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of CS&E, BIT 2025-26 94 Observations □ When reasoning embeddings lie close to or overlap with the input embedding clusters, the Blind Agent demonstrates strong semantic preservation, indicating that reasoning remains faithful to the original audio–text input.

□ In several cases, the reasoning embeddings shift moderately away from the input clusters, indicating semantic drift.

□ Larger spatial separation between ellipsoids corresponds to greater abstraction or restructuring of semantic content rather than outright reasoning failure.

Interpretation The observed semantic drift reflects adaptive reinterpretation, not instability.

When input semantics are ambiguous, weakly aligned, or underspecified, the Blind Agent reorganizes its reasoning trajectory to maintain conceptual clarity.

This behavior aligns with the system's neuroplasticity-inspired design, where reasoning adapts dynamically to sensory constraints.

Figure 84 3D PCA alignment between input embeddings and Blind-Agent reasoning Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 95 embeddings 8142 2D Semantic Alignment Projection To further quantify the relationship between input semantics and Blind Agent reasoning, a two-dimensional PCA projection was employed.

In this representation □ The x-axis corresponds to the projected input centroid.

□ The y-axis corresponds to the projected reasoning centroid.

Figure 85 presents the resulting 2D semantic alignment plot, where each point represents a test input processed by the Blind Agent.

Observations □ Points lying near the diagonal reference line indicate strong semantic alignment between input and reasoning embeddings.

□ Minor vertical deviations from the diagonal correspond to controlled elaboration, where the Blind Agent enriches or refines the explanation beyond the literal input.

□ Larger deviations indicate adaptive reinterpretation, typically occurring for abstract, incomplete, or semantically weak inputs.

Interpretation The 2D projection provides an interpretable summary of Blind Agent behavior □ The agent maintains high semantic alignment for well-defined and linguistically coherent inputs.

□ Deviations are systematic rather than random, confirming intentional reasoning adaptation rather than noise or instability.

□ The Blind Agent prioritizes semantic clarity and explanatory completeness, even when strict embedding alignment is not achievable.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 96 Figure 85 2D semantic alignment between inputs and Blind Agent reasoning outputs

Summary of Blind Agent Behaviour The combined 3D and 2D analyses demonstrate that □ The Blind Agent exhibits robust semantic consistency under aligned audio–text conditions.

□ Observed semantic drift is adaptive and meaningful, not indicative of reasoning failure.

□ Reasoning trajectories reorganize dynamically based on linguistic structure, conceptual density, and input ambiguity.

Overall, these findings validate the Blind Agent’s role as a stable, linguistically grounded reasoning component within the proposed neuro-adaptive multimodal system.

The agent’s behavior closely reflects the intended neuroplasticity-inspired cognitive design, balancing semantic preservation with adaptive reasoning under constrained sensory conditions.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 97 82 Snapshots □ Home page Figure 86 Home Page Home page of the Neuro-Adaptive Multi-Agent System, which serves as the primary entry point for users to initiate multimodal reasoning.

The interface introduces the system’s neuroplasticity-inspired design and provides direct access to launch the Normal, Blind, and Mute agent workflows.

□ Multimodal Interaction Page Figure 87 Multimodal Interaction Page Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 98 Figure 87 illustrates the main interaction interface of the Neuro-Adaptive Multi-Agent Intelligence system, where users can provide text, image, and audio inputs and selectively invoke the Normal, Blind, or Mute agents.

The interface also supports 3D input visualization, enabling users to observe multimodal semantic alignment before initiating agent-level reasoning.

□ Multimodal Input Illustration Figure 88 Multimodal Input Illustration Figure 88 demonstrates an example of multimodal input provided to the system, where the user supplies a text query, an image, and an audio sample simultaneously.

This snapshot illustrates how heterogeneous sensory inputs are collected and prepared for semantic encoding and subsequent neuro-adaptive agent reasoning.

□ 3D Input Modality Visualization Page This snapshot visualizes the encoded image, audio, and text inputs in a shared 3D semantic space, helping users understand how different modalities are positioned and related prior to reasoning.

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 99 Figure 89 Input Modality Visualization After visualization, the system displays the extracted image caption, audio transcript, and original text input, providing transparent insight into how raw inputs are interpreted internally.

Figure 810 Input Transcripts Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 100 □ Reasoning Embedding Space Visualization This snapshot shows the 3D reasoning embedding space where the reasoning outputs of the Normal, Blind, and Mute agents are projected using PCA. Each agent’s reasoning is represented as a point in the shared semantic space, with distances illustrating how closely the agents’ reasoning aligns or diverges under the same input conditions.

Figure 811 Reasoning Embedding Space □ Output Page This snapshot illustrates the unified reasoning interface

where the Normal, Blind, and Mute agents execute their respective reasoning pipelines in parallel.

The page displays modality-aware outputs (text, audio, or both).

Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 101 Figure 812 Output CHAPTER 9 APPLICATIONS & CONCLUSION Dept.

of CS&E, BIT 2025-26 102 CHAPTER 9 APPLICATIONS & CONCLUSION 91 Applications The proposed Neuro-Adaptive Multi-Agent System provides a flexible framework for intelligent reasoning in environments where multimodal inputs may be incomplete, noisy, or unavailable.

By dynamically adapting reasoning strategies based on available modalities, the system can be applied to several real-world domains that require robustness, interpretability, and adaptive intelligence.

911 Healthcare Systems The system can support healthcare applications that involve heterogeneous data sources by enabling adaptive multimodal reasoning.

Potential applications include ☐ Clinical decision support, where medical images, clinician voice notes, and textual patient records are jointly analyzed to assist diagnosis.

☐ Patient monitoring systems, in which reasoning adapts based on available inputs such as speech patterns, textual symptoms, or visual indicators.

☐ Rehabilitation and assistive care, where adaptive feedback mechanisms can support patients with sensory or cognitive impairments by tailoring explanations to their capabilities.

912 Assistive Technologies The neuro-adaptive design makes the system particularly suitable for assistive AI solutions ☐ Support for visually impaired users, where the Blind Agent compensates for missing visual input using audio and text reasoning.

☐ Support for speech- or hearing-impaired users, where the Mute Agent relies on image and text inputs to generate structured explanations.

☐ General sensory compensation systems, enabling intelligent interaction even when one or more input modalities are unavailable.

913 Autonomous and Intelligent Robotic Systems In robotic and autonomous systems, the framework can enhance reasoning under uncertain sensing conditions ☐ Robust perception, where missing or noisy sensor inputs are compensated using Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 103 learned semantic representations.

☐ Adaptive decision-making, allowing robots to reformulate reasoning when certain sensors fail.

914 Human–AI Collaboration Systems The proposed framework enhances human–AI collaboration by enabling

☐ Context-aware virtual assistants for education and training environments.

☐ Customer service agents capable of understanding user intent through text, speech, or visual inputs.

☐ Industrial automation systems, where adaptive reasoning improves safety, efficiency, and explainability in human-in-the-loop settings.

915 Smart Edge Devices and IoT Systems The system is well-suited for deployment in smart edge and embedded environments, including ☐ Real-time multimodal processing on edge devices with limited computational resources.

☐ IoT-based monitoring systems, such as smart homes and wearable technologies.

☐ Surveillance and embedded intelligence platforms, where adaptive reasoning is required under partial or degraded sensory input.

92 Conclusion This project presented a Neuro-Adaptive Multi-Agent Intelligence System inspired by the principle of human neuroplasticity.

The system demonstrates how artificial agents can adapt their reasoning behavior based on the availability of sensory inputs rather than relying on rigid multimodal assumptions.

By integrating three specialized agents Normal, Blind, and Mute the framework enables robust and interpretable reasoning under different modality constraints.

The proposed architecture combines ☐ Multimodal feature extraction and embedding alignment ☐ A unified low-dimensional latent representation ☐ Memory-augmented retrieval for contextual reasoning ☐ Agent-specific adaptive reasoning strategies ☐ Interactive explanation generation through a web-based interface Neuro Adaptive AI Agents Dept.

of CS&E, BIT 2025-26 104 Experimental evaluation using unit testing, integrated testing, and PCA-based visualization confirms that the system dynamically reorganizes its reasoning pathways in response to modality

availability.

The results show consistent semantic alignment, effective modality compensation, and coherent agent-level reasoning behavior.

Overall, the work demonstrates a practical and interpretable approach to adaptive multimodal intelligence and provides a foundation for future research in neuro-inspired AI systems.

93 Future Scope of the Work Although the proposed system achieves adaptive multimodal reasoning, several extensions can further enhance its capabilities.

931 Reinforcement-Based Adaptation Future versions may integrate reinforcement learning mechanisms to □ Adjust reasoning strategies based on feedback and success metrics □ Learn optimal modality prioritization over time □ Adapt explanation depth according to user interaction patterns 932 Enhanced Neuroplastic Modeling Long-term adaptability can be improved by incorporating □ Neuro-symbolic reasoning layers □ Dynamic memory consolidation and pruning mechanisms □ Improved abstraction and reasoning continuity across sessions 933 Collaborative Multi-Agent Expansion The framework can be extended to support □ Multiple agents collaborating and sharing learned knowledge □ Distributed reasoning across agent groups □ Cooperative problem solving in complex environments 934 Cross-Domain Validation Future work may evaluate the system across additional domains such as □ Medical diagnostics Neuro Adaptive AI Agents Dept. of CS&E, BIT 2025-26 105 □ Autonomous navigation □ Personalized education and cognitive assistance 935 Edge and Embedded Deployment Optimized versions of the system may be deployed on □ Embedded platforms and robotic hardware □ Edge-AI devices with low latency requirements □ Real-time decision-making systems operating under sensory constraints 936 Temporal Knowledge Update Mechanism A continuous learning loop can be introduced in which □ Knowledge from previous interactions informs future reasoning □ Cross-modal validation refines belief states □ Memory and reasoning evolve incrementally over time The conceptual design of this temporal knowledge update loop is illustrated in Figure 91.

Figure 91 Temporal reinforcement and knowledge update loop.

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Dept of CS&E, BIT 2025-26 108 ANNEXURE A GLOSSARY □ Agent Manager A control module responsible for selecting, coordinating, and combining outputs from multiple specialized reasoning agents based on input modality availability and confidence.

- Audio Encoder A neural network model that converts raw audio or speech features into a low-dimensional latent representation used for similarity matching and reasoning.
- Automatic Speech Recognition (ASR) A system that converts spoken language into textual form, enabling audio inputs to participate in multimodal reasoning.
- Blind Agent A specialized neuro-adaptive agent that performs reasoning using only audio and text modalities, mimicking the cognitive behavior of a visually impaired individual.
- Cognitive Memory A long-term memory module that stores compressed semantic summaries, embeddings, and contextual metadata derived from previous reasoning experiences.
- Cosine Similarity A mathematical measure used to determine semantic similarity between two embedding vectors based on their angular distance.
- Dimensionality Reduction The process of reducing high-dimensional embeddings (eg, 384D) into lower dimensions (eg, 3D) for visualization and analysis.
- Emotion Recognition The process of identifying emotional context from text, audio, or facial expressions to enhance reasoning relevance and importance scoring.
- Embedding A numerical vector representation of data such as text, audio, or images that captures semantic meaning in a latent space.
- Flask Framework A lightweight Python web framework used to build the interactive user interface for multimodal input submission and reasoning visualization.
- Image Captioning A computer vision technique that generates descriptive text for images, enabling visual information to participate in language-based reasoning.
- Key–Value Memory Representation A memory structure where compact latent keys (15D) are used for fast similarity matching and higher-dimensional values (384D) store rich semantic content.

Dept of CS&E, BIT 2025-26 109 □ Latent Space A continuous vector space in which embeddings of different modalities are aligned to enable similarity computation and reasoning.

- Memory Retriever A module that retrieves the most relevant stored memories using weighted similarity measures combining semantic similarity, latent key similarity, and modality boosting.
- Modal Compensation A neuro-adaptive mechanism where available modalities compensate for missing or degraded inputs during reasoning.
- Multimodal Learning A learning paradigm that integrates multiple data modalities such as text, audio, and images to achieve richer understanding.
- Multimodal Fusion The process of integrating information from multiple modalities into a unified semantic representation for reasoning.
- Mute Agent A specialized neuro-adaptive agent that performs reasoning using image and text modalities only, simulating conditions where auditory input is unavailable.
- Neuro-Adaptive System An artificial intelligence system inspired by biological neuroplasticity that dynamically reorganizes reasoning pathways based on sensory availability and context.
- Neuroplasticity The ability of a system biological or artificial to adapt its internal structure and reasoning pathways in response to experience or sensory changes.
- Normal Agent A multimodal reasoning agent that utilizes image, audio, and text inputs simultaneously to generate the most comprehensive interpretation.
- PCA (Principal Component Analysis) A statistical technique used to reduce high-dimensional embeddings into lower dimensions for visualization and analysis of semantic relationships.
- Reasoning Fusion Engine A decision-making module that integrates outputs from all agents, memory retrieval results, and cross-modal semantics into a unified reasoning vector.
- Semantic Drift A measurable change in semantic representation between input embeddings and reasoning embeddings, reflecting adaptive reinterpretation.
- Semantic Embedding A vector representation capturing the meaning of content in a form suitable for similarity computation and reasoning.
- Sentence-BERT (SBERT) A transformer-based model used to generate 384- dimensional semantic

embeddings for text, captions, and transcriptions.

□ Similarity Scoring A weighted computation combining semantic similarity, Dept.

of CS&E, BIT 2025-26 110 latent key similarity, keyword overlap, and modality boosting to rank memory relevance.

□ Speech Emotion Recognition The process of identifying emotional cues from speech to enhance contextual understanding.

□ Top-N Retrieval A memory access strategy that retrieves the N most relevant stored memories based on similarity scores.

□ Voice Activity Detection (VAD) A technique used to detect the presence of speech in audio signals before transcription.

□ Weighted Similarity Function A scoring mechanism that combines multiple similarity measures with adjustable weights to optimize retrieval accuracy.